
FEDERAL PURCHASE DESCRIPTION

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR 40 FEET- 34 PASSENGER

BATTERY- ELECTRIC

NPS-ZION

1.0 Scope and Classification

1.1 Scope

The buses will be used in Zion National Park (Zion). Zion's shuttle system was implemented in the year 2000 to address growing visitation and increasing resource impacts, and to improve visitor experience. Park staff and visitors alike have benefitted from the reduced congestion, emissions, and noise levels within Zion as a result of the shuttle system. The existing shuttle fleet is propane powered and at the end of its service life. The goal of this battery electric bus purchase is to further enhance the visitor experience and resource protection through additional reductions in exhaust emissions, fuel use, and noise levels.

This specification defines requirements for heavy duty, 40 foot-long, minimum 34 seated passengers, perimeter seating configuration, battery-electric buses. The bus shall be low floor design, be a maximum of 102 inches wide, have a left-hand driver location, and be fully ADA-compliant.

In addition, the bus shall have a minimum expected life of 12 years/500,000 miles and be intended for the widest possible spectrum of passengers, including children, adults, the elderly, and persons with disabilities. The bus shall meet other design goals, including long cycle life, low life-cycle cost, safety, maintainability, durability, and robust diagnostics.

This Purchase Description does not include all varieties of the commodity indicated by the title, but is intended to cover those vehicles generally acquired by the Government. A selection of optional additional systems and equipment are included for agency's operational related needs.

The buses shall be capable of performing in the following environment. The Zion Canyon route will operate on a 16-mile loop, 8-miles each way continuously for 15-hours per day, 7 days per week, March through November, several weekends in February, and several days during Christmas/New Years. On average the Canyon buses make 8-11 loops per day. (Approximately 128 to 176 miles per day plus standby time when not en route.) The Springdale route will operate on a 5.8-mile loop, 2.9-miles each way continuously for 15-hours per day March through November. On average the Springdale buses make 21-30 loops per day. The buses do not typically operate December through February. (Approximately 122 to 174 miles per day plus standby time when not en route.)

Zion's shuttle service is operated along paved roads in a high desert environment that has relatively low humidity and temperature changes of more than 30 degrees Fahrenheit in a day. Ambient

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temperatures during the operational season range from 40 to 120 degrees Fahrenheit (F) (4 to 49 degrees Celsius (C)). During shoulder season and off seasons, the buses may be exposed to freezing temperatures both while in use and while parked.

Average operational speed is 10-12 miles per hour, with a top speed of 32 miles per hour.

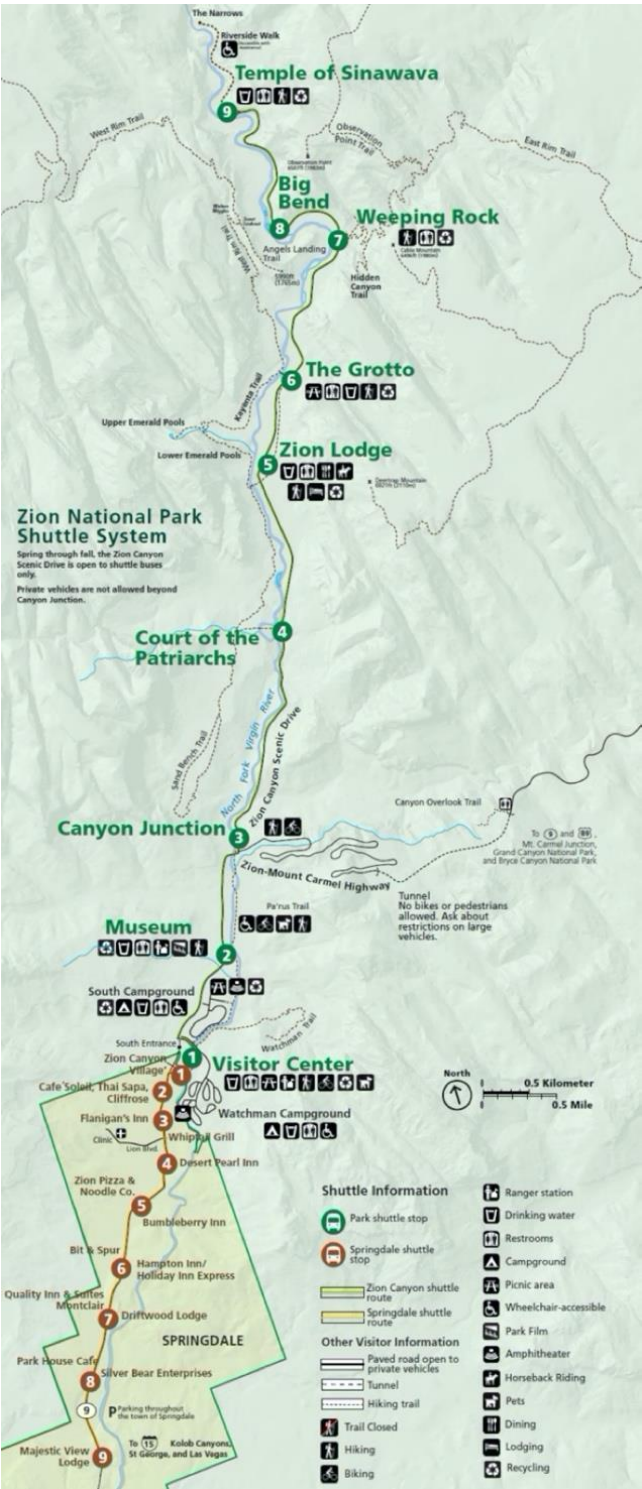
Range

Key Performance Parameters, the ability of NPS Zion to meet their mission:

achieve a range of 200 miles with a full charging cycle based upon the defined routes within the park

The Zion Canyon route takes passengers from the visitor center to the Temple of Sinawava and back.

The Springdale route takes passengers from the town to the visitor center area and back.



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Zion's bus maintenance facility currently operates on 3-phase, 480/277 Volt electrical service. Charging infrastructure has been initiated in the outdoor bus parking area near the bus maintenance facility. Offeror shall communicate full electrical service requirements and forecasted load demands for thirty (30) battery electric buses to the National Park Service. The National Park Service will supply the information to the regional power provider for expected service upgrades.

1.2 Classification

The individual bus configuration classifications are referred to by a Standard Item Number (SIN). The SINs covered by this specification are as follows:

377H-Bus-Transit-Heavy Duty-Low Floor-40 Ft.- 34 Passenger-Battery Electric Zion

2.0 Applicable Documents

The vehicles shall conform to the latest revision of the following:

Federal Regulations and Standards:

Federal Motor Carrier Safety Regulations (FMCSR), 49CFR 393

Federal Motor Vehicle Safety Standards (FMVSS), 49CFR 571

Americans with Disabilities Act, 49 CFR 38

FTA Docket 90-A, October 20, 1993

Application for copies of DOT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. They may also be accessed on the Internet through GPO Access at <http://www.access.gpo.gov>.

The body of this Purchase Description (PD) contains requirements relative to the following standards and recommended practices, and the offeror's proposed vehicles shall conform to all such requirements.

Society of Automotive Engineers (SAE) Standards and Recommended Practices:

- J10 Automotive and Off-Highway Air Brake Reservoir Performance and Identification Requirements
- J287 Driver Hand Control Reach
- J516 Hydraulic Hose Fittings
- J517 Hydraulic Hose
- J534 Lubrication fittings
- J553 Circuit Breakers
- J678 Speedometers and Tachometers—Automotive
- J941 Motor Vehicle Drivers' Eye Locations
- J1050 Describing and Measuring the Driver's Field of View
- J1052 Motor Vehicle Driver and Passenger Head Position
- J1131 Performance Requirements for SAE J844 Nonmetallic Tubing and Fitting Assemblies Used in Automotive Air Brake Systems

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J1149 Metallic Air Brake Tubing and Pipe
J1522 Truck Driver Stomach Position
J1625 Heavy Duty Circuit Breakers
J1708 Serial Data Communications Between Microcomputer Systems in Heavy-Duty Vehicle Applications
J163 Low tension Wiring and Cable Terminals and Splice Clips
J198 Windshield Wiper Systems- Trucks, Buses, and Multi-Purpose Vehicles
J382 Windshield Defrosting Systems Performance Requirement-Trucks, Buses, and Multi-Purpose Vehicles
J593 Backup Lamps (Reversing Lamps)
J673 Automotive Safety Glasses
J844 Nonmetallic Air Brake System Tubing
J683 Tire Chain Clearance
J686 Motor Vehicle License Plates
J833 Human Physical Dimensions
J941 Motor Vehicle Drivers Eye Location
J994 Alarm—Backup—Electric Laboratory Performance Testing
J1127 Battery Cable
J1128 Low Tension Primary Cable
J1273 Recommended Practices for Hydraulic Hose Assemblies
J1292 Automobile, Truck, Truck-Tractor-Trailer, and Motor Coach Wiring
J1654 High Voltage Primary Cable
J1673 High Voltage Automotive Wiring Assembly
J1766 Electric, Fuel Cell, and Hybrid Electric Vehicle Crash Integrity Testing
J1772 Electric Vehicle Conductive Charge Coupling
J1939 Serial Control and Communications Network
J2464 Electric and Hybrid Electric Vehicle Rechargeable Energy System (RESS) Safety and Abuse Testing
J2929 Electric and Hybrid Electric Vehicle Propulsion Battery System Safety Standard
J2293/2- Energy Transfer Systems for Electric Vehicles- Communication Requirements and Network

Application for copies of SAE publications should be addressed to:

SAE World Headquarters <http://www.sae.org/>
400 Commonwealth Drive
Warrendale, PA 15096-0001

Industry Standards

National Fire Protection Association (NFPA) <http://www.nfpa.org/>

NFPA 70 (National Electrical Code-NEC)

NFPA standards may be obtained from:

National Fire Protection Association
1 Batterymarch Park

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Quincy, MA 02169-7471

American Public Transportation Association (APTA) <http://www.apta.com/Pages/default.aspx>

APTA recommendations may be obtained from:

APTA
1666 K Street, NW
Suite 1100
Washington, DC 20006
Telephone: (202) 496-4800

National Institute for Automotive Service Excellence (ASE) <http://ase.com/>

ASE requirements may be obtained from:

National Institute for Automotive Service Excellence
101 Blue Seal Drive, S.E.
Suite 101
Leesburg, VA 20175
Phone 703-669-6600

Underwriters Laboratory, Inc. (UL) <http://www.ul.com/>

UL 2202- Standard for Safety of Electric Vehicle (EV) Charging System Equipment
UL 2231- Standard for Safety of Personnel Protection Systems for EV Supply Circuits
UL 2251- Standard for Safety of Plugs, Receptacles, and Couplers for EVs
UL 2580-Standard for Batteries for Use in Electric Vehicles
UL 1642- Standard for Lithium Batteries
UL 2594- Standard for Electrical Vehicle Supply Equipment
UL1254- Standard for Safety for Pre-Engineered Dry Chemical Extinguishing System Units

Underwriters Laboratory standards may be obtained from:

2600 N.W. Lake Rd.
Camas, WA 98607-8542
Telephone: 1.877.UL.HELPS (1.877.854.3577) or
<http://ulstandardsinfonet.ul.com/catalog/stdscatframe.html>

3.0 Overall Requirements

The bus shall comply with all applicable federal, state, and local regulations. These shall include, but not be limited to, the Federal Americans with Disabilities Act (ADA), Federal Motor Vehicle Safety Standards and Regulations 49 CFR 571 (FMVSS), 49 CFR 393 (FMCSR), National Fire Protection Association (NFPA), United States Environmental Protection Agency (USEPA), as well as state and local accessibility, safety, and security requirements.

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The offeror shall certify to the Government that the vehicle platform complies with 49 CFR, Bus Testing Rule. The offeror shall test the bus platform proposed at the Altoona (PA) Testing Facility and provide copies of all testing results, forms, and documentation as part of the proposal offer. The bus vehicle platform proposed shall be tested in accordance with the Altoona testing category requirements for heavy-duty large buses (minimum service life of 12 years or 500,000 miles). If the bus platform proposed by the offeror has already been tested successfully at the Altoona Testing Facility, the offeror shall provide verification of the successful testing with the offer.

The offeror shall explicitly and completely identify any and all exceptions taken to the requirements of this specification. Drawings, literature, and any other information submitted with an offer shall not constitute a stated or implied exception unless specifically identified in writing as an exception and accepted and implemented by an amendment to the solicitation or modification to the contract. Any exception deemed acceptable by the Government shall only apply to the specific item, requirement, etc. cited, and shall not extend to any other requirement.

Materials shall be as specified herein. When materials are not specified, the vehicles and all parts thereof shall be furnished to provide the intended function, durability, safety, and maintain good long-term appearance. All materials shall be new, shall be suitable for the intended purpose, and shall be free of any characteristics or defects in material and workmanship, which may affect the performance, function, durability, and serviceability of the finished vehicle, or detract from its appearance. The Government reserves the right to make the final determination of the suitability of all components and their arrangement on and in the vehicle.

The offeror responding to this solicitation shall be a bus body and chassis OEM- Original Equipment Manufacturer. The offeror shall have demonstrated the commercial manufacture and sale of low floor battery electric transit buses of designs equal to or of greater complexity than the requirements herein. Commercial order and sales data shall be submitted with the offer.

The offeror is encouraged to submit with its offer value engineering information (refer to Federal Acquisition Regulation [FAR] Part 48) in areas of vehicle design, construction, space utilization, etc. which will improve the quality, function, durability, and performance of the vehicle and/or provide a cost savings to the Government. Adoption of these submissions will be determined by the Government and will only be implemented by an amendment to the solicitation or modification to the contract.

The offeror shall submit with its offer the following:

- Interior plan view of the proposed bus, including seating arrangement, wheelchair locations, and dimensions verifying compliance with the requirements herein.
- Exterior front, rear, and side view with overall dimensions.

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- Estimated curb weight and maximum GVW with total seated and standee passengers plus driver. The total number of seated and standee passengers based on this definition of gross load shall be furnished with the offer.
- Estimated weight distribution at front and rear axles at maximum GVW.

Description of life-cycle testing procedures used to validate the life of the energy storage system at the specified charging rates, charge durations, and specified ambient temperatures and operating profiles. The offeror shall include documented results of life cycle testing and certification of battery life cycle testing by an independent testing agency.

The offeror shall ensure that the application and installation of major bus subcomponents and systems are compliant with subcomponent vendors' requirements and recommendations. Components used in the vehicles shall be of heavy-duty design and proven in transit service.

The offeror shall provide route simulations and test reports that demonstrate the bus shall perform as specified within this document.

3.0.1 Dimensions

3.0.1.1 Physical Size

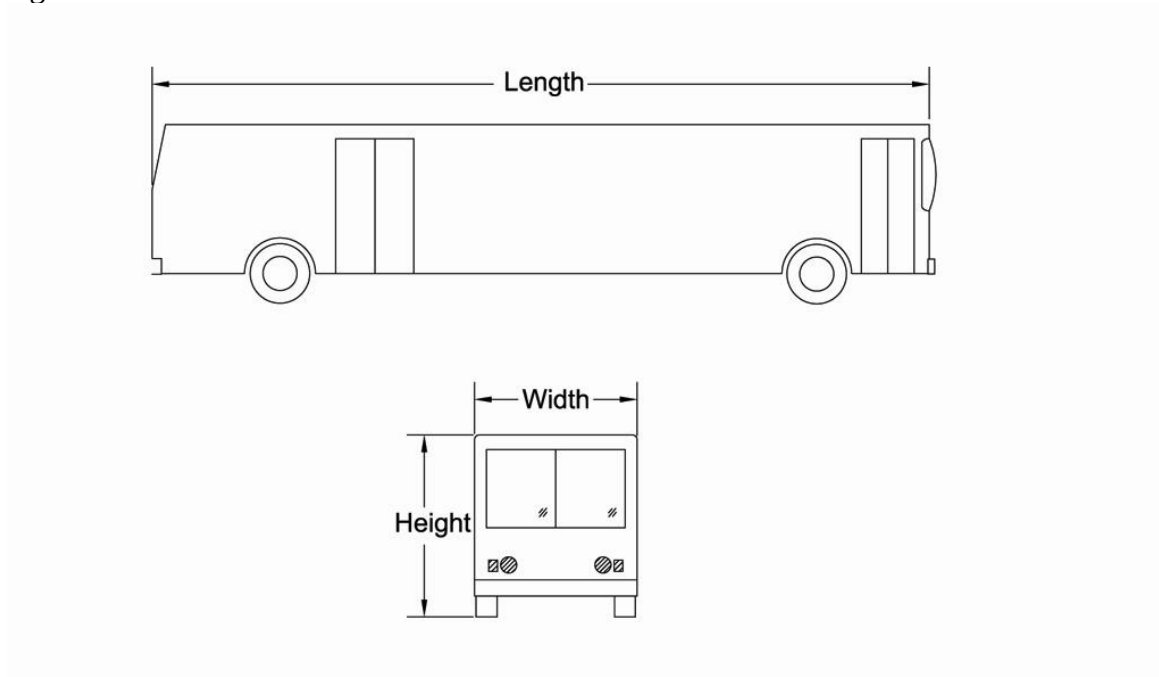
With the exception of exterior mirrors, marker and signal lights, bumpers, fender skirts, and rub rail, the buses shall have the following overall dimensions (as illustrated in Figure 1 below).

Bus Overall Dimensions:

- | | |
|---------------------------------------|-----------------------------|
| • Length (excluding bumpers) | 40 feet (± 8 inches) |
| • Width (exterior, excluding mirrors) | 102 inches (+ 0/- 1 inches) |
| • Overall Height | 140 inches (max.) |

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Figure 1: Bus Overall Dimensions



3.0.1.2 Underbody Clearance

All components under the bus shall be protected from any impact with foreign objects.

- Wheel Area Clearance - Wheel area clearance, shall be 8 inches (minimum) for parts fixed to the bus body and 5 inches (minimum) for parts that move vertically with the axles.
- Ramp Clearances - Approach angle shall be 8 degrees (minimum). Breakover angle shall be 8 degrees (minimum). Departure angle shall be 8 degrees (minimum). Any encroachment into the approach or departure angle area shall encounter a structural member before any operating component or subsystem.
- Axle Zone Clearance - Axle zone clearance, which is the projected area between tires and wheels on the same axial centerline, shall be as great as possible within the confines of overall vehicle width.
- Ground Clearance - Ground clearance shall be 9 inches (minimum), except within the axle zone and wheel area.

3.0.1.3 Weight

3.0.1.3.1 Curb Weight

Curb weight of the bus shall be minimized to the greatest extent practicable while maximizing its integrity and durability.

Weight distribution shall be such that no axle shall be loaded past its GAWR at maximum gross vehicle weight with the gross load weight distribution in accordance with all applicable State and Federal requirements.

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The Contractor shall submit to the Government during vehicle delivery, a certified weight slip for the curb weight of each axle and total vehicle weight.

3.0.1.3.2 GVWR and Capacity

The Gross Vehicle Weight Rating of the bus shall exceed the curb weight of the bus plus the capacity of passengers and driver. Total passenger capacity shall be the number of people (combination seated/standing) considering a minimum of 34 passenger seats that can be carried with the offeror's maximum rated (gross axle weight rating) front and rear axle, using 150 pounds/passenger and driver. See also Section 4 Definitions for the gross load. The GVW and axle loadings shall not exceed the GVWR and front and rear GAWR at maximum capacity. For purposes of calculations, the standee passengers may be equally distributed on the bus in the area allowed for standees.

The offeror shall furnish, with the offer, the maximum front and rear GAWR, the GVWR, and the maximum number of passengers (seated and standee) the bus can carry without exceeding the GAWR and GVWR at the 150 pounds /passenger and driver requirement.

3.0.1.3.3 Passenger Capacity Decal

Instruction decals shall be applied above the exterior front door and to the interior forward bulkhead visible to all passengers and driver. The lettering shall be not less than 1-1/2 inch in height and in strong contrast to the interior background color. The message can be in a single line or two lines and shall state "Maximum Passenger Capacity XX" or "Capacity XX Passengers". The two X's shall represent the total seated and standee passenger capacity.

3.0.1.4 Life Expectancy

The bus shall be designed to operate in transit service for at least 12 years and/or 500,000 miles.

3.0.1.5 Maintainability Requirements

Routine, scheduled maintenance or inspection tasks as specified by the Contractor shall require an ASE (National Institute for Automotive Service Excellence) technician skill level of 3M or less (target). To facilitate ease of maintenance, the vehicle design shall maximize the use of line replaceable units (LRUs). All LRUs shall allow expedient replacement.

Readily accessible test ports shall be provided for routinely checked functions on-board the proposed vehicle (such as hydraulic, pneumatic, charge-air, and cooling system(s)).

3.0.1.6 Material Selection/Interchangeability

The bus shall be designed using commercially available (off-the-shelf) components and materials to the greatest extent possible. For economy in maintenance, it is essential that all parts and units be arranged so that rapid assembly and disassembly will be possible for the vehicles being provided. The dimensions of all parts, unless particularly specified, shall be in accordance with current standards of the Society of Automotive Engineers (SAE), or the metric equivalent. All units or parts not specified shall be Contractor's standard units or parts and shall conform in material, design and workmanship to industry standards and shall meet or exceed all federal and state motor vehicle safety standards.

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Components with identical functions shall be interchangeable to the extent possible. These components shall include passenger lamps (and components), window hardware, trim, interior components, seat assemblies and other components considered to be consumable. Components with non-identical functions shall not be, or appear to be, interchangeable. All parts shall be new and in no case will used, reconditioned, or obsolete parts be accepted. No advantages shall be taken by the manufacturer in the omission of any parts or details that make the vehicles complete and ready for service, even though such parts or details are not mentioned in these specifications.

Components and replacement parts shall be available for a minimum of 10 years after delivery.

3.0.1.7 Ambient Operation Requirements

The bus shall be engineered for operation in an ambient temperature range of -30 degrees to +115 degrees Fahrenheit (at relative humidity levels between 5 and 100 percent, and at altitudes up to 7,000 feet above sea level). Speed, gradeability, and acceleration performance requirements shall ideally be met at, or corrected to, 77 degrees Fahrenheit, 29.31 inches Hg, dry air per SAE J1995.

3.0.1.8 Elderly and Disabled Passenger Accessibility

The Contractor shall comply with all applicable Federal requirements defined in the Americans with Disabilities Act, 49 CFR Part 38, and all state and local regulations regarding mobility-impaired persons. Local regulations are defined as those below the state level.

3.0.2 Audible Noise Emission Control

Since noise levels are of a primary concern to the Government, the Government strongly prefers that each vehicle have a low level of exterior/interior noise and, as a design target, that each vehicle be significantly quieter than stated herein. If the offeror has optional sound deadening capability, the maximum sound deadening materials and insulation shall be furnished. The combination of inner and outer panels, plus any material used between them, and motor compartment insulation shall provide sufficient sound insulation so that sound levels do not exceed the values in sections below.

3.0.2.1 Interior Noise Emissions

The interior noise shall be tested in accordance with Altoona Bus Testing interior noise test procedure.

A. With the left exterior of the bus exposed to a “white” noise minimum level of 80 dB(A), and all accessories “off”, the noise transmitted to the interior of the bus shall not exceed 65 dB(A).

B. Under full throttle acceleration from 0-35 MPH, and all accessories operating, the bus generated interior noise levels shall not exceed the values below:

- 80 dB(A) at any passenger seat location
- 78 dB(A) at the driver’s seat

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3.0.2.2 Exterior Noise Emissions

The exterior noise shall be tested in accordance with Altoona Bus Testing exterior noise test procedure.

Airborne noise generated by the bus and measured from either side shall not exceed 80 dB(A) under full power acceleration when operated at or below 35 mph at curb weight and just prior to transmission upshift. The maximum noise level generated by the bus pulling away from a stop at full power shall not exceed 80 dB(A). The bus-generated noise, with bus stationary, from either side at low or high idle, with accessories and air conditioning “on” shall not exceed 65 dB(A). All noise readings shall be taken 50 feet from, and perpendicular to, the centerline of the bus. The pull away test shall begin with the front bumper even with the microphone. The stationary idle test shall be conducted with the rear bumper even with the microphone.

3.0.3 Back Up Alarm

The bus shall be furnished with an audible alarm that sounds whenever the ignition switch is in the “on” position and the drive is in reverse gear.

3.0.4 Back Up Camera

The bus shall be furnished with a camera that provides an image on a video screen on the driver’s console when the ignition switch is in the “on” position and the drive is in reverse gear.

The bus shall also be furnished with interior cameras located at the rear entry doors connected to the video screen. The driver shall select be able to select the rear entry door camera inputs to the video screen via momentary switch(es).

3.0.5 Electromagnetic Radiation (EMI)

Electrical and electronic sub-systems and components on all bus shall comply with FCC regulations.

Electrical and electronic sub-systems on the vehicles shall not be affected by external sources of RFI/EMI. This includes, but is not limited to, radio and TV transmission, portable electronic devices including computers in the vicinity of or onboard the bus, AC or DC power lines and RFI/EMI emissions from other vehicles.

Electronic components that are subject to RFI/EMI shall have shielded power and data cabling. Deviations from this requirement shall require Government approval. On-board equipment can include but is not limited to automatic passenger counters, two-way radio, electronic motor and transmission controls, automatic vehicle locating equipment, and destination signs and associated wiring.

If one vehicle and/or component is susceptible to RFI/EMI, it will be assumed that all vehicles suffer the same defect. Design corrections shall be made on a fleet basis. The Government reserves the right to review and approve proposed solutions.

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Upon the request of the Government, the Contractor shall submit test data or other evidence that all of the above requirements are met.

3.0.6 Responsibility for Design

The contractor shall assume total and overall responsibility for the design and satisfactory operation of the vehicle and vehicle subsystems or component parts. The contractor's responsibility includes, but is not limited to, ensuring that the design and manufacture of the vehicle and its component parts are appropriate, coordinated, compatible, and perform correctly throughout the life of the vehicle (either together or individually).

3.0.7 OEM Requirements and Recommendations

The proposed bus is an assembly of components that may be sourced from several different original equipment manufacturers (OEMs) or suppliers. The Contractor shall obtain and adhere to the requirements and recommendations of the OEMs for the application and installation of components that are installed. In the absence of specific OEM requirements and/or recommendations, the Contractor shall obtain the OEM's approval for their installation. Components utilized in the vehicle shall be of heavy-duty design and proven in transit service.

When delivered the bus shall be in a ready-to-use condition. Lubricants and any other fluids utilized shall be of the type specified by the OEMs and installed at the levels recommended by the OEMs. Energy system liquid coolant (if required by the OEM) shall be biodegradable propylene glycol, Chevron Delo ELC PG, or equal. Ethylene glycol is prohibited.

3.0.8 Technology Upgrades

The offered bus vehicle shall blend service-proven design principles and equipment with the latest technological advancements without sacrificing reliability, maintainability, safety or performance.

The successful Contractor shall provide the Government with written notification (including a complete description of upgrade/advancement, trade-off analysis, and commercial implications) of viable technology upgrades and advancements throughout contract duration.

3.1 Chassis

3.1.1 Top Speed

The maximum top speed shall be governed to not exceed 65 MPH.

3.1.2 Electric Drive

The electric drive system shall provide power to enable the proposed buses to meet the defined acceleration, top speed, gradeability, operate all accessories, while still maintaining passenger comfort and providing a smooth ride. Electric drive input torque rating shall exceed motor output torque. The drive system shall be rated to operate at the GVWR of the bus.

The electronic drive shall have on-board diagnostic capabilities, be able to monitor functions, store and time stamp out-of-parameter conditions in memory and communicate faults and vital conditions to service personnel. The electronic drive shall contain built-in protection software

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to guard against severe damage. A diagnostic reader device connector port, suitably protected against dirt and moisture, shall be provided in the driver's area.

3.1.2.1 Bus Interface

The system controller shall be the point of interface to the bus electrical platform and the preferred communication path shall be serial data (J-1939, CAN). The system controller shall also be capable of communication to a variety of on-board device controllers, including but not limited to, ABS, drive motor(s), and other.

3.1.2.2 Driver Interface

Driver interface shall include selection of desired direction, throttle, and visual indicators of faults. Alarms requiring immediate or indicating planned automatic shutdown shall also be audible. All interfaces shall be ergonomically designed.

3.1.2.3 Energy Storage System

The Energy Storage System (ESS) shall be of a commercial design. The ESS shall be designed, sized, and selected to ensure that the bus performance specifications, compatibility with charging, and other related requirements are met or exceeded, bearing in mind cost benefit and reliability variables as they relate to the characteristics of the different battery types. The power source for the vehicle shall be derived from established battery technology purpose built for bus propulsion applications that has a field-proven track record of safe, reliable, and durable operation in similar traction applications.

The energy storage system shall be composed of batteries along with associated power electronics interface and controls, diagnostic systems, and environment controls of a commercial design for transit duty. This system shall be designed to provide a load-leveling function in the propulsion system. The energy storage devices used, and their arrangement shall be selected and sized to meet bus performance specifications and design goals, including: long cycle life, low life-cycle cost, safety, maintainability, durability, and simple, robust diagnostics.

The energy storage system shall monitor and control the operating conditions within each module, including voltage, current, and temperature. This system shall include over-current and over-temperature protection features that disconnect current to and from the energy storage modules in the event of an over- temperature or over-current condition, and de-rating performance (limits current) in the case of an over temperature condition.

The energy storage system shall be capable of being replaced and upgraded.

Offerors shall include complete descriptions of all life-cycle testing procedures used to validate the life of batteries used at the proposed charging rates, charge durations, and expected ambient temperatures and operating profiles. Offerors shall include documented results of life cycle testing and certification of battery life cycle testing by independent testing agency.

3.1.2.4 Thermal Management

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If necessary, to maintain thermal management of the energy storage system under the ambient operating conditions required herein, the energy storage system shall be furnished with a self-contained, manufacturer approved, refrigerated cooling system dedicated only to the thermal management of the energy storage system. A warning light shall be supplied on the driver's panel indicating abnormal operational conditions (i.e.: excessive battery compartment temperature, loss of airflow, loss of power) within the energy storage device cooling system. This cooling system shall be rated to provide the required thermal management under the ambient operating conditions required herein and shall be manufactured and installed in compliance with the specifications and requirements of the manufacturer.

3.1.2.5 Electrical Safety System

A high voltage interlock monitoring and activation system shall be used to determine intentional or unintentional opening of high voltage circuits or enclosures and automatically neutralize all devices capable of or producing high voltage power. This system shall be fully compliant with SAE J1673, and other applicable standards, and shall be active when the master run switch is in the OFF/CL/ID position and shall monitor all components and associated cabling within the system. When the master run switch is in the RUN/night position, the monitoring system shall provide an audible and visual warning on the driver's panel and provide for automatic shutdown of the high voltage system if a shutdown event occurs. The high voltage system includes all bus platforms operating above 50 VDC or 25 VAC and any variety of modulated currents to/from power electronic converters.

An electrical isolation fault detecting system shall monitor the level of isolation between the high voltage system (positive and negative) and the bus chassis and be fully compliant with SAE J1766, and all other applicable standards. The designed safety characteristics shall be supplemented by the use of properly labeled and located safety tags and plates in accordance to SAE J1654 and J1673.

The High-Voltage (HV) battery system design shall be a holistic integrated system with safety set as the absolute top priority. The HV cables shall utilize industry standard shielding and insulation, cabling shall be routed and secured to avoid chafing with other cables and components. The HV battery system shall also require a low-voltage interlock circuit that will automatically de-energizing the HV system if a HV box is opened. The vehicle controls system shall monitor all major components inside each battery pack as well as provide an overall battery system controller and an overall bus system controller.

3.1.2.6 Energy Storage System Charging

The energy storage unit shall be isolated from the passenger compartment so as to prevent any intrusion of liquids or gasses relating to the energy storage unit into the passenger compartment. Charging inlets shall be provided on both the street and curb sides of the bus.

All lights on the bus shall be deactivated during charging in compliance with NPS Zion dark skies policies, shall provide zero illumination during charging.

3.2 Propulsion System

3.2.1 Propulsion System

The bus shall be powered by a battery electric propulsion system. Function and operation of the bus shall be transparent to the bus driver and passengers. The Contractor shall assure that

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the bus structure can successfully accept the installation of the propulsion system and be operated on the stated duty-cycle for a period of 12 years without a structural failure. At a minimum, the propulsion system shall comply with applicable local, state, and/or federal emissions and useful life requirements, as a zero-emission bus.

3.2.1.1 Propulsion System Service

The propulsion system shall be arranged so that accessibility for all routine maintenance is assured. No special tools, other than dollies and hoists, shall be required to remove the propulsion system or any subsystems. Contractor shall provide all specialty tools and diagnostic equipment required for maintaining the Propulsion System in accordance with Special Tools List.

3.2.2 Primary Traction Motor

The bus shall have a permanent magnet (PM) traction motor . The PM motor shall be designed for 300,000 miles of operation without failure or deterioration in performance.

3.2.3 Propulsion System Controller (PSC)

The PSC regulates energy flow throughout system components in order to provide motive performance and accessory loads, as applicable, while maintaining critical system parameters (e.g., voltages, currents, temperatures, etc.) within specified operating ranges. The controller shall monitor and process inputs and execute outputs as appropriate to control the operation of all propulsion system components. The overall propulsion system and PSC shall include and manage support systems such as, steering, air, HVAC, defroster, etc.

The PSC shall be the hub for all propulsion system device to device communication, to include traction motors, energy storage, charging equipment and power switching electronics, and interface to other vehicle systems via SAE J1939. The PSC shall provide the following functionality:

- Storage of the application file necessary to execute propulsion system commands

- Storage of the buses data file generated on a day-to-day basis, to include:

 - Incidents and alarms

 - Monitoring and diagnostics information

 - Wireless communications

- The system shall include current / power sensors at strategic locations throughout the propulsion system.

- The system shall include sensors at strategic locations, such as, temperature, voltage, pressure.

- The system shall protect the traction motor(s) against progressive damage. The system shall monitor conditions critical for safe operation and automatically de-rate power and/or speed.

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The system shall include a sub-system capable of monitoring the level of connectivity between all propulsion components and associated cabling/connectors to the buses chassis and low (12/24) voltage systems to insure isolation. The energy storage module shall have automatic means/devices of disconnect and one manual capable of interrupting the positive and negative connections within the module enclosure and rated for disconnect at maximum current.

The system shall have an interlock that prevents engagement when the charger is connected to the traction battery.

The PSC shall be equipped with an electronically controlled management system, compatible with multiplex wiring systems and either 12-or-24volt electrical systems.

3.2.4 Power Electronics/Inverter

The PSC shall execute instructions and system commands to the propulsion system components via a power electronics inverter. Circuitry for this device (s) shall include all necessary fuses/breakers such that the conductors, components and bus are adequately protected and safe. Connection points shall be keyed/identified such that mismatch is not physically possible. Reconnecting the subject connector (s) will not automatically restore the connection to the energy storage module; a system reset will be required.

3.2.5 Energy Storage System

The Energy Storage System (ESS) shall be of a commercial design capable of operating in a transit environment.

The primary charging of the energy storage system shall be provided from a stationary charging station conforming to all UL standards and SAE J1772 CCS. The energy storage system shall also make use of regenerative braking. The Energy Storage System shall comply with UN/DOT 38.3 or UL2580 or UL1642 requirements for lithium batteries or similar standards for non-lithium batteries.

The Contractor shall deliver the buses with an installed, fully functioning ESS. The ESS shall be fully formed, installed and tested in accordance with the manufacturer's recommended practices. The ESS design, including containers, module bracing systems, thermal-management systems, battery-management systems, watering/venting systems, interconnections, fusing, and traction-controller and charger interfaces shall be completely described in the proposal. The proposal shall include a detailed analysis of expected battery performance in the Design Operating Profile.

The charge cycle and cycle life shall be stated in the proposal and a life-cycle cost analysis of the proposed battery system in the specified application shall be provided.

Offerors shall include complete descriptions of all life-cycle testing procedures used to validate the life of batteries used this application at the proposed charging rates, charge durations, and expected ambient temperatures and operating profiles. Offerors shall include documented results of life cycle testing.

3.2.5.1 Energy Storage System Safety

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The Energy Storage System shall be installed to optimize interior space and vehicle weight distribution. The ESS shall be load distributed within the bus to equalize weight between the wheels on the same axles and to achieve appropriate weight distribution between axles so as not to adversely affect handling of the bus. The ESS shall be positioned to maintain safety and not be impacted in side-impact crash scenarios.

3.2.6 Cooling System

The cooling system shall be of sufficient size to maintain power electronics and the ESS at safe continuous operating temperatures, in accordance with the power electronics system component manufacturers' cooling system requirements and recommendations.

The cooling system shall be automatically controlled under normally encountered operating and charging conditions. Thermal management shall be continuously monitored during all periods of charge and discharge with appropriate safety interlocks installed to react to adverse conditions as stated in SAE-J1772.

Air intakes shall be properly positioned and configured to minimize the intake of water, road dust, and debris and shall be adequately filtered.

In the event of a failure of the cooling system while charging, the charge system shall be disabled and a visual alert shall be activated on the dashboard, the reset of which shall require the deliberate action of maintenance personnel. In the event of a failure of the cooling system during bus operation, an audible and visual alert shall be activated on the dashboard, the reset of which shall require the deliberate action of maintenance personnel. In the event of a fire onboard a bus, fans shall be automatically turned off.

3.2.7 Regenerative Braking

The bus shall have a regenerative braking system to aid in the reduction of wear on the brakes and to help extend the range of the vehicle through energy recapture. The vehicle shall employ regenerative braking as the accelerator pedal is completely released.

3.2.7.1 Regenerative Brake Management

The system shall include a means of maintaining dynamic braking (braking retardation) that is designed to prevent overcharging of the batteries. This same feature may be a component of the overall liquid cooling system loop and offer a means of supplementing heat for use at the main HVAC heater core and/or defroster.

3.3 Power Steering

A power steering system shall be provided. The steering gear shall be either axle-mounted or frame-mounted. The steering column shall have telescoping and tilt column adjustments. The steering columns shall be an integral type with the number and length of flexible lines minimized.

Fatigue life of all mechanical steering components shall exceed the service life of the bus. The steering system shall be considered as part of the basic body structure.

Turning effort:

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Steering torque applied by the driver shall not exceed 10 foot-pounds with the front wheels straight ahead to turned 10 degrees. Steering torque may increase to 70 foot-pounds when the wheels are approaching the steering stops. Steering effort shall be measured with the bus at GVWR, stopped with the brakes released, and at normal idling speed on clean, dry, level, commercial asphalt pavement and tires inflated to recommended pressure. Power steering failure shall not result in loss of steering control. With the bus in operation, the steering effort shall not exceed 55 pounds at the steering wheel rim, and perceived free play in the steering system shall not materially increase as a result of power assist failure. Gearing shall require no more than 7 turns of the steering wheel lock-to-lock.

Caster angle shall be set to provide a tendency for the return of the front wheels to the straight position with minimal assistance from the driver.

Turning Radius:

The outside body corner turning radius shall not exceed 44 feet with the bus at seated load weight (SLW).

3.4 Air Compressor

An electrically driven air compressor and air system shall be furnished that meets the following requirements:

Minimum 13 CFM

The compressor discharge hose shall be SAE 100R14A PTFE, stainless steel braid covered hose routed to a point isolated from compressor vibration and movement (i.e. to a bulkhead). Zinc plated steel, or stainless steel, tubing is acceptable if run to a component on the assembly. Copper tubing of any type is not acceptable as discharge tubing directly off the compressor discharge head.

3.5 Fire Safety

The bus is designed and manufactured in accordance with all applicable fire safety and smoke emission regulations. These provisions include the use of fire-retardant/low-smoke materials, fire detection systems, bulkheads and facilitation of passenger evacuation.

No fire suppression system shall be required so long as the Battery Management System is capable of thermally monitoring the internal temperature with redundant sensors internal to the battery packs. If the temperatures become high enough to affect performance, the Battery Management System shall be required to de-rate power until the temperature is reduced. If the temperatures were to continue rising, the control system shall be required to disable the vehicle.

3.6 Fluid Lines, Fittings and Clamps, and Charge Air Pipework

All fluid lines and air pipework shall be rigidly supported to prevent chafing damage, fatigue failures, and tension strain. All support clamps shall be rubber covered to prevent chafing and cutting. All hose clamps shall be constant torque type.

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Cooling piping shall be stainless steel or brass tubing.

Cooling, oil and hydraulic lines shall be compatible with the fluid they carry. The lines shall also be compatible with potentially damaging elements of the surrounding environment including heat. Lines shall be capable of withstanding maximum system pressures.

Flexible lines shall be kept at a minimum and shall be as short as practicable. Flexible lines shall be routed or shielded so that failure of a line shall not allow fuel or oil to spray or drain onto any component operable above the auto-ignition temperature of the fluid.

Flexible hoses and fluid lines shall be individually supported and shall not touch one another or any part of the bus.

Charge air pipework and fittings shall be designed to minimize air restrictions and leaks. Pipework shall be as short as possible and the number of bends minimized. Bend elbows shall have smooth contours and shall be maximized to meet the pressure drop and temperature rise requirements of the manufacturers. The cross section of all charge air pipework shall not be less than the cross section of the intake manifold inlet. Any changes in pipework diameter shall be gradual to ensure a smooth passage of air and to minimize restrictions. Pipework shall be routed away from exhaust manifolds and other heat sources and shall be insulated and shielded from these sources to protect inlet air from heat radiation.

Charge air pipework shall be seamless, constructed of stainless steel, aluminum, or aluminized steel. Connections between all charge air pipework sections shall be sealed with a short Section of reinforced hose and stainless steel, constant tension clamps that provide a complete 360 degree seal.

3.7 Suspension

3.7.1 General

All axle suspensions shall be of the pneumatic type and shall have a load rating equal to or exceeding that of the axles. The basic elements of the suspension system shall last the life of the bus without major overhaul or replacement. Suspension beams, weldments, and structural members shall be considered as parts of the basic body structure. Items such as bushings and air springs shall be easily and quickly replaceable by a 3M mechanic in 30 minutes or less. Suspension pivots shall be replaceable. Bushings shall be permanently lubricated or require no lubrication. Adjustment points shall be minimized and shall not be subject to a loss of adjustment in service. Front and rear axle sway bars may be utilized if necessary.

3.7.2 Damping

Vertical damping of the suspension system shall be accomplished by hydraulic shock absorbers mounted to the suspension arms or axles and attached to an appropriate location on the chassis. Damping shall be sufficient to control coach motion to 3 cycles or less after disturbance. Shock absorbers shall maintain their effectiveness for at least 50,000 miles of the service life of the bus. Each unit shall be replaceable by a 2M mechanic in less than 1 hour.

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3.7.3 Lubrication

All elements of steering, suspension, and drive systems requiring scheduled lubrication shall be provided with grease fittings conforming to SAE Standard J534. These fittings shall be located for ease of inspection and shall be accessible with a standard grease gun without flexible hose end from a pit or with a bus on a hoist. Each component that requires lubrication shall have its own grease fitting with relief path. Lubricant specified shall be standard for all elements on the bus serviced by standard fittings.

3.7.4 Travel

Elastomeric bumpers shall be provided at the limit of jounce travel. Rebound travel may be limited by elastomeric bumpers or hydraulically within the shock absorbers. Suspensions shall incorporate appropriate devices for automatic height control so that regardless of load the bus height relative to the centerline of the wheels does not change more than $\pm \frac{1}{2}$ inch at any point.

3.7.5 Kneeling

A front and rear kneeling system shall lower the entrances of the bus during loading or unloading operations regardless of load up to GVWR. Release of the control switch shall completely stop lowering motion and hold height of the bus at that position. Upward direction of the switch will allow the system to go to floor height without the driver having to hold the switch up.

Brake and throttle interlocks shall prevent movement when the bus is kneeled. The kneeling control shall be disabled when the bus is in motion. The bus shall kneel at a maximum rate of 2 inches per second at essentially a constant rate. After kneeling, the bus shall rise within 4 seconds to a height permitting the bus to resume service and shall rise to the correct operating height within 7 seconds regardless of load up to GVWR.

An indicator visible to the driver shall be illuminated until the bus is raised to a height adequate for safe street travel. An audible warning alarm will sound simultaneously with the operation of the kneeler to alert passengers and bystanders. A warning light mounted near the curbside of the front door, minimum 2.5 inches diameter, amber lens shall be provided that will blink when the kneel feature is activated. Kneeling shall not be operational while the wheelchair ramp is deployed or in operation.

3.8 Axles

Front and rear axles shall have a minimum load rating that exceeds the actual load on the axles when the bus is loaded to the GVWR and shall operate for 300,000 miles without repairs. The front and rear axle ratings shall be the offeror's highest available rating to maximize the GVWR and the number of seated and standee passengers based on the gross load definition in Section 4 Definitions. The axle gearing shall be easily accessible for lubrication.

3.9 Brakes

The brakes shall be of the full-air internal expanding S-Cam type or disc type and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.52 and FMVSS 571.121. Brake linings shall have extra service transit linings. Brake linings/pads shall be fabricated of non-asbestos material.

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Brakes shall be self-adjusting. Any metallic air brake tubing shall comply with SAE J1149.

3.9.1 Parking Brake

The parking brake shall be a spring-operated system, actuated by a valve that exhausts compressed air to apply the brakes. The parking brake may be manually enabled when the air pressure is at the operating level per FMVSS 121. An emergency brake release shall be provided to release the brakes in the event of automatic emergency brake application. The parking brake valve button shall pop out when air pressure drops below requirements of FMVSS 121. The driver shall be able to manually depress and hold down the emergency brake release valve to release the brakes and maneuver the bus to safety. Once the driver releases the emergency brake release valve, the brakes shall engage to hold the bus in place.

3.10 Air System

3.10.1 General

The bus air system shall operate the air-powered accessories and the braking system with reserve capacity. A new bus shall not leak down more than 5 psi as indicted on the instrument panel mounted air gauges, within 15 minutes from the point of governor cut-off.

Provision shall be made to apply shop air to the bus air systems using a standard tire inflation type valve. A quick disconnect fitting shall be easily accessible and located in the rear compartment and near the front bumper area for towing. Retained caps, where approved for use by the OEM, shall be installed to protect fittings against dirt and moisture when not in use. Air for the compressor shall be filtered an air cleaner system. The air system shall be protected by a pressure relief valve and shall be equipped with check valve and pressure protection valves to assure partial operation in case of line failures.

3.10.2 Air Lines and Fittings

Air lines subjected to temperatures over 200 Degrees F shall be SAE100R14A PTFE, stainless steel braid covered hose, and all other airlines shall be nylon tubing manufactured in accordance with SAE J844 with performance meeting the requirements of SAE J1131. Nylon tubing shall be installed in accordance with the following color-coding standards:

- Green: Primary brakes and supply
- Red: Secondary brakes
- Brown: Parking brake
- Blue: Suspension
- Yellow: Compressor Governor

- Black: Accessories.

Nylon lines may be grouped and shall be continuously supported and prevented from any movement, flexing, tension strain, chafing, and vibration. They shall not touch one another or any part of the bus except for the supporting grommets. Flexible lines shall be supported at 30 inch intervals or less. Grommets of bulkhead fittings shall protect the air lines at all points where they pass through understructure components.

3.10.3 Air Reservoirs

All air reservoirs shall meet the requirements of FMVSS Standard 121 and SAE Standard J10. The wet or “ping” tank shall be furnished with automatic, heated, drain valve. All other air reservoirs shall be equipped with manual drain valves, unless mounting locations allow draining via the wet tank. The manual drain valves shall be located such that they may be drained from the ground at the side of the bus. The system shall be furnished with a pressure relief apparatus equipped with check valve. Major structural members shall be provided to protect these valves from road hazards. Reservoirs shall be sloped toward the drain valve to assure that condensate is eliminated by the valve. All air reservoirs and tanks shall have a non-corrosive treatment applied to them prior to installation on the bus vehicle (subject to Government review and approval).

3.10.4 Air Dryer

An air dryer with replaceable desiccant cartridge shall be furnished on all air systems and installed per compressor and air dryer manufacturer’s recommendations. The dryer shall be located with ease of access for inspection and replacement.

3.11 Hydraulic Systems

Any accessory may be driven hydraulically. The hydraulic system shall demonstrate a mean time between repairs in excess of 50,000 miles. Hydraulic system service tasks shall be minimized and scheduled no more frequently than those of other major bus systems. All elements of the hydraulic system shall be easily accessible for service or unit replacement.

3.11.1 Hydraulic systems shall meet the following requirements:

Hydraulic systems shall meet the following requirements:

Hydraulic system hoses, fittings, pressures, and flow rates.

Hydraulic hoses shall be rubber covered double wire braid reinforced and comply with SAE 100R2, type A or AT, or 100R9, type A or AT, or single wire braid reinforced hose complying with SAE 100R5 or SAE J517. The working pressure of the hose shall exceed the pressure setting of the relief valve. Hoses shall be sized such that the maximum velocity of hydraulic fluid in the hose does not exceed the following:

Fluid velocity in suction lines shall not exceed 4 ft./sec.

Fluid velocity in discharge lines shall not exceed 25 ft./sec

All hoses shall be installed in accordance with the requirements and recommendations of SAE J1273

System working pressure shall not exceed 3500 PSI.

High pressure hydraulic hose fittings shall comply with the requirements of SAE J516 for permanently attached (crimped) fittings with JIC 37° flare. Field replaceable type fittings are acceptable on low pressure return and suction hoses. Forged steel hydraulic

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adapters shall be used. Cast steel fittings are acceptable only in low pressure circuits, such as suction lines.

Any other hydraulic system shall be designed with adequate cooling (reservoir dwell time or auxiliary cooler) such that the maximum oil temperature does not exceed 200 Degrees F.

3.12 Odometer, Hub Mounted

Each bus shall be supplied with a hub-odometer.

3.13 Tires and Wheels

3.13.1 Tires

Tires shall be suitable for the conditions of transit service and sustained operation at the maximum speed capability of the bus. Load on any tire at GVWR shall not exceed the tire supplier's rating.

The Contractor shall provide "plain" valve system caps with each mounted tire. No valve stem tool will be permitted on the valve stem cap.

3.13.2 Wheels

Wheels shall be drop-forged aluminum and hub piloted. The wheels shall be polished on both sides. All wheels shall be interchangeable and shall be removable without a puller. Wheels shall be of the ventilated type with 10 studs, radial approved, reverse mount, and made for tubeless tires. Wheels shall be compatible with tires in size and load-carrying capacity.

Front wheels and tires shall be balanced as an assembly per SAE J1986. Front wheels and tires shall be spin balanced as an assembly using appropriate weight.

The contractor shall furnish one fully balanced spare wheel/tire assembly, shipped loose with each bus.

3.14 Service Requirements

3.14.1 General

The traction motor bay shall be arranged so that accessibility for all routine maintenance is assured. Two (2) 3M mechanics shall be able to remove and replace the drive motor assembly in less than twelve (12) total combined man-hours.

All major components, accessories, and/or any other component requiring service or replacement shall be mounted in an accessible location. Each bus shall be designed to facilitate the disassembly, re-assembly, servicing or maintenance by use of tools and items that are normally available as commercial standard items.

3.14.2 Fillers

Fluid, oil and coolant filler caps shall be properly labeled. All fillers shall be accessible with standard funnels, pour spouts, and automatic dispensing equipment.

3.14.3 Accessories

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Wherever possible, all battery driven accessories, such as alternator, power steering pump, and air compressor, shall be modular and unit-mounted for quick removal and repair. Accessory drive systems shall operate without failure or unscheduled adjustment for 50,000 miles on the design operating profile.

3.15 Body

3.15.1 Design

The bus shall be of integral body and understructure construction and shall have a clean, smooth, simple design and uniform appearance. The exterior and body features, including grilles and louvers, shall be shaped to facilitate cleaning by automatic bus washers without causing damage to either the bus vehicle body or the washer brushes. Water and dirt shall not be retained in or on a body feature to freeze or bleed out onto the bus after leaving the washer. The body and windows shall be sealed to prevent leaking of air, dust, or water under normal operating conditions and during cleaning in automatic bus washers for the service life of the bus.

Accumulation on any windows of the bus of spray and splash generated by the bus wheels on a wet road shall be minimized. Heated exterior rearview mirrors shall be furnished.

3.15.2 Materials and Construction

All body materials shall be selected and the body fabricated for simplified replacement and repair, to reduce maintenance, extend durability, and provide consistency of appearance throughout the service life of the bus. Add-on devices and trim, where necessary, shall be minimized and integrated into the basic design.

3.15.3 Crashworthiness

The bus vehicle body and roof structure shall withstand a static load equal to 150 percent of the curb weight evenly distributed on the roof with no greater than a 6-inch reduction in any interior dimension. Under such a load, windows shall not open and will remain in place.

At any point (excluding doorways), the bus vehicle shall withstand a 25-mph impact by a 4,000-pound automobile along either side of the bus with no more than 3 inches of permanent structural damage at seated passenger hip height. Sharp edges or protrusions in the bus vehicle interior shall not result from this impact.

Exterior panels located below 35 inches from ground level shall withstand a static load of 2,000 pounds applied perpendicular to the bus by a pad no larger than 5 inches square. Deformation that prevents installation of new exterior panels to restore the original appearance of the bus vehicle shall not result from the application of this load.

The Offeror shall furnish to the Government evidence of the crashworthiness capabilities of the proposed bus.

3.15.4 Fire Protection

The bus shall be designed and manufactured in accordance with all applicable fire safety and smoke emission regulations. These provisions include the use of fire-retardant/low-smoke materials, fire detection systems, bulkheads and facilitation of passenger evacuation. All

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materials used in the construction of the passenger compartment of the bus are in accordance with the Recommended Fire Safety Practices defined in FMVSS 302.

A certified fire extinguisher, standard size ABC rated dry chemical, carbon dioxide, minimum 2.3 kg (5 pound) , with dial type indicator, in quick-release brackets, shall be installed on each bus vehicle by the contractor. Proposed placement shall permit effective access by the bus vehicle driver and passengers in the event of fire while minimizing potential for vandalism or inadvertent tampering. Proposed placement within the bus vehicle shall be defined by the contractor and will be approved by the Government.

3.15.5 Resonance and Vibration

To minimize audible, visible, or sensible resonant vibrations during normal service, all structure, body, and panel-bending mode frequencies, including vertical, lateral, and torsional modes, shall be sufficiently removed from all primary excitation frequencies.

3.15.6 Corrosion Protection

The bus chassis structure shall resist corrosion or deterioration from atmospheric conditions and road salts for a period of 12 years or 500,000 miles whichever comes first. It shall maintain structural integrity and maintain nearly original appearance throughout its service life, provided that it is maintained by the Government in accordance with the procedures specified in the contractor's service manual. With the exception of periodically inspecting the visible coatings applied to prevent corrosion and reapplying these coatings in limited spots, the contractor shall not require the complete reapplication of corrosion compounds over the life of the bus.

All exposed surfaces and the interior surfaces of tubing and other enclosed members shall be corrosion resistant. All sub-assembly components requiring painting shall be primed and painted before installing into their next assembly.

All materials that are not inherently corrosion resistant shall be protected with corrosion-resistant coatings. All joints and connections of dissimilar metals shall be corrosion-resistant and shall be protected from galvanic corrosion. Representative samples of all materials and connections shall withstand a 2-week (336-hour) salt spray test in accordance with ASTM Procedure B-117 with no structural detrimental effects to normally visible surfaces, and no weight loss of over 1 percent.

3.15.7 Insulation

Any insulation material used between the inner and outer panels shall minimize the entry and/or retention of moisture. Insulation properties shall be unimpaired during the service life of the bus. Any insulation material used inside the motor compartment shall not absorb or retain oils or water and shall be designed to prevent casual damage that may occur during maintenance operations.

The combination of inner and outer panels on the sides, roof, wheel wells and ends of the bus, and any material used between these panels, shall provide a thermal insulation sufficient to

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meet the interior temperature requirements. The bus body shall be thoroughly sealed so that the driver or passengers cannot feel drafts during normal operations with the passenger doors closed.

3.16 Structure

3.16.1 Strength and Fatigue Life

The basic structure shall be designed so that fatigue damage will not occur during the service life of the bus. The vehicle structure shall withstand impact and inertial loads due to street travel as representative in transit service throughout its service life, without permanent deformation or damage.

3.16.2 Hoisting

The bus axles and/or jacking plates shall accommodate the lifting pads of a 2-post hoist system. Jacking plates, if used as hoisting pads, shall be designed to prevent the bus from falling off the hoist. Other pads or the bus structure shall support the bus on jack stands independent of the hoist. The hoisting and jacking of a bus vehicle shall not create an over-capacity condition to any of the axles or air bags that may cause damage.

3.16.3 Jacking

It shall be possible to safely jack up the bus, at curb weight, with a common 10-ton floor jack with or without special adapter, when a tire or dual wheel set is completely flat and the bus is on a level, hard surface (without crawling under any portion of the bus). Jacking from a single point shall permit raising the bus sufficiently high to remove and reinstall a wheel and tire assembly. Jacking pads located on the axle or suspension near the wheels shall permit easy and safe jacking with the flat tire or dual wheel set on a run-up block not wider than a single tire. Jacking and changing any one tire shall be completed by a serviceman in less than 30 minutes from the time a bus is approached. The bus shall withstand such jacking at any one or any combination of wheel locations without permanent deformation or damage.

Jacking pads shall be painted safety yellow or orange for easy identification.

3.16.4 Towing

Towing devices shall be provided on the front end only of the bus. Each towing device shall withstand, without deformation, tension loads up to 1.2 times the curb weight of the bus within 20 degrees of the longitudinal axis of the bus.

Front towing devices shall allow attachment of adapters for a rigid tow bar and shall permit lifting and towing of the bus, at curb weight, until the front wheels are off the ground.

3.16.5 Flooring

3.16.5.1 Design

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The floor shall be a continuous flat plane, unless interior steps are provided. Where the floor meets the walls of the bus, as well as other vertical surfaces, such as platform risers and wheel wells, the surface edges shall completely sealed with an appropriate sealant. The floor shall be sloped within ADA limitations in door entry thresholds to facilitate adequate drainage.

3.16.5.2 Construction

The floor shall consist of the subfloor and the floor covering. The floor, as assembled, including the sealer, attachments and covering shall be waterproof, non-hygroscopic, and resistant to mold growth. The subfloor shall be composite, plywood is prohibited. The edges of the floor shall not be exposed to tire water spray. The rear step/transition area between the upper and lower decks shall be composite flooring.

3.16.5.3 Strength

The floor deck may be integral with the basic structure or mounted on the structure securely to prevent chafing or horizontal movement and designed to last the life of the bus. The use of adhesives to secure the floor to the structure shall be allowed only in combination with the use of bolt or screw fasteners and its effectiveness shall last throughout life of the bus. All floor fasteners shall be secured and protected from corrosion for the service life of the bus. The floor deck shall be reinforced as needed to support passenger loads. At GVWR, the floor shall have an elastic deflection of no more than 0.60 inches from the normal plane. The floor shall withstand the application of 2.5 times gross load weight without permanent detrimental deformation. Floor, with coverings applied, shall withstand a static load of at least 150 pounds applied through the flat end of a 2 inch-diameter rod, with 1/32-inch radius, without permanent visible deformation.

3.17 Wheel Housings

3.17.1 Design

Sufficient clearance and air circulation shall be provided around the tires, wheels, and brakes to preclude overheating when the bus is operating on the design operating profile.

Wheel housings shall be adequately reinforced where seat pedestals are installed to ensure satisfactory performance over the life of vehicle.

3.17.2 Construction

Wheel housings, as installed and trimmed, shall withstand impacts of a two inch steel ball with at least 200 LBs-FT of energy. The housings shall have sufficient insulation to minimize tire and road noise.

Wheel housings shall be designed to maximize the interior aisle width and preclude water from being trapped between the flooring and the top of the wheel housing. Wheel housings shall be equipped with liners and/or sealed to prevent water and dirt from entering or collecting in the frame or structure, causing corrosion.

3.17.3 Clearance

Sufficient clearance and air circulation shall be provided around tires, wheels, and brakes to prevent overheating when the bus is operating. Wheels and tires shall be removable when a

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bus vehicle is lifted by the axle or suspension, even with the air bags depleted. Tire chain clearance shall be provided on all driven wheels.

3.18 Exterior

3.18.1 Strength and Installation

Body materials shall be selected and the body fabricated to reduce maintenance, extend durability and provide consistency of appearance throughout the service life of the bus. Detailing shall be kept simple, and add-on devices and trim are minimized and integrated into the basic design

3.18.2 Rain Gutters

Rain gutters shall be provided to prevent water flowing from the roof onto side windows, doors, and exterior mirrors. When the bus vehicle accelerates or decelerates, the gutters shall not drain onto the windshield, or driver's side window, or into the door boarding area. Cross sections of the gutters shall be adequate for proper operation and gutters shall extend a minimum of two inches beyond openings.

3.18.3 License Plates

Provisions shall be made to mount standard size U.S. license plates per SAE J686 on the front and rear of the bus. To the extent practicable, license plates shall not allow a toehold or handhold for unauthorized riders.

3.18.4 Bicycle Rack

A front bicycle rack shall be furnished and installed, Sportsworks model Velo-Porter, or equal, demountable rack for transporting three (3) bicycles on the front of each bus. The bike rack shall not interfere with attaching towing equipment or accessing towing connectors and it shall not obstruct the driver's field of vision.

3.19 Exterior Finish, Color, and Graphics

The exterior and roof shall be painted single light shade determined by NPS Zion at no later than 180 days ARO. The contractor shall provide additional paint finish as required plus exterior graphics, including large-scale text reading "Zion Shuttle" and the NPS arrowhead artwork on both sides and on the back of the bus. Specifications and vector file artwork will be provided by the NPS no later than 60 days ARO.

All exterior surfaces shall be smooth and free of wrinkles and dents. Exterior surfaces to be painted shall be properly prepared as required by the paint system supplier, prior to application of paint to assure a proper bond between the basic surface and successive coats of original paint for the service life of the bus. Drilled holes and cutouts in exterior surfaces shall be made prior to cleaning, priming and painting to prevent corrosion. Any paint shall be applied prior to installation of exterior lights, windows, mirrors and other items that are applied to the exterior of the bus. Body filler materials may be used for surface dressing, but not for repair of damaged or improperly fitted panels.

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Depending upon the primer/topcoat system selected, painted surfaces shall be cleaned with solvent, chemical, or mechanical methods meeting the paint manufacturer's recommendations.

Paint used shall be a rust inhibitive primer and polyurethane topcoat or equal.

Paint or gelcoat shall be applied smoothly and evenly with the finished surface free of dirt and the following other imperfections:

- A. Blisters or bubbles appearing in the topcoat film.
- B. Chips, scratches, or gouges of the surface finish.
- C. Cracks in the film.
- D. Craters where finish failed to cover due to surface contamination.
- E. Overspray.
- F. Peeling
- G. Runs or sags from excessive flow and failure to adhere uniformly to the surface.
- H. Chemical stains and water spots.
- I. Orange peel

All dents, roughness, or other surface imperfections shall be corrected prior to the application of the priming coat. Primer and finish paint shall be supplied as a system by a single manufacturer. Preparation for finish application shall follow the finish manufacturer's recommendations. As a minimum, prior to finish application, surfaces shall be cleaned to remove all traces of contamination.

Finish shall be applied evenly, and the finished surface shall be free of dirt, runs, "orange peel" or other imperfections. At least two coats of paint shall be applied, with appropriate surface preparation between coats. Touch-up finish shall be identical in all respects to the original finish.

Prior to assembly, all low-alloy steel areas shall be painted with one coat of an approved primer followed by one coat of an approved sealer to prevent rusting.

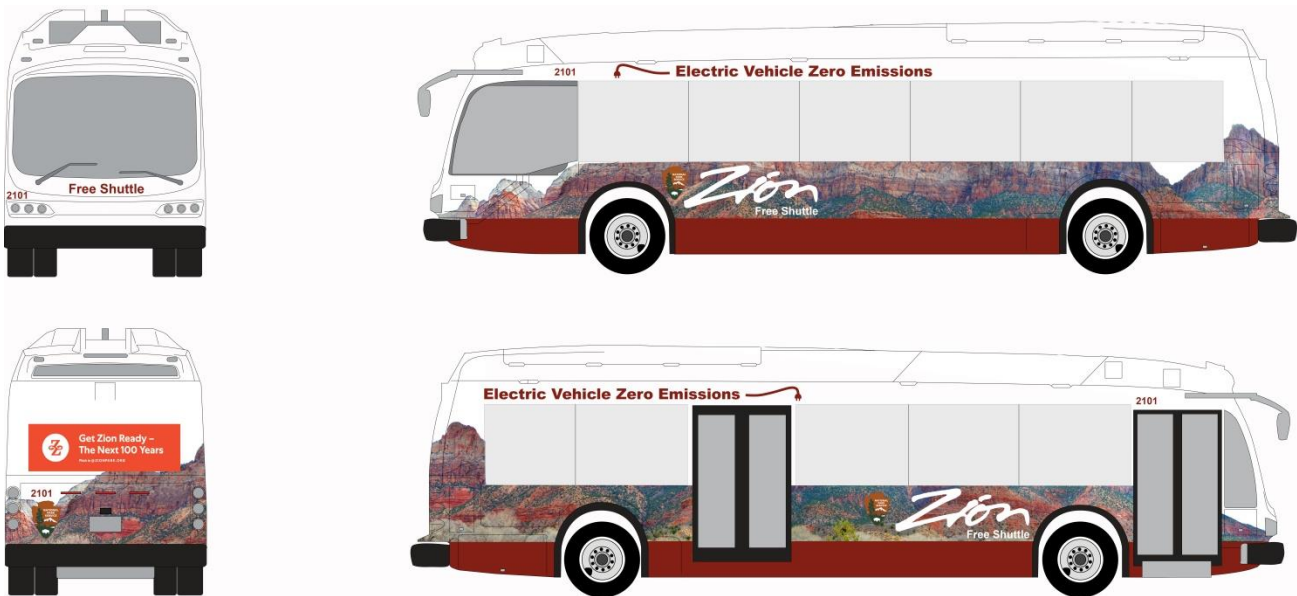
Each bus shall be furnished with an exterior graphics package in accordance with the following requirements:

- Each bus shall be furnished with an exterior multi-color graphics system designed for long term use in extreme conditions. The graphics system shall have a minimum 4.5 year warranty against fading, cracking, and peeling.
- The graphics system shall be a 3M Scotchprint® Matched Component System compatible with the final graphic design(s).

The successful offeror shall provide Adobe Illustrator templates of the offered bus configurations at the Post Award Conference (PAC). NPS Zion shall provide artwork, no later 180 days ARO.

The successful offeror shall provide a full size, hard copy, printed wrap material sample (minimum size 3'x3') of the bus wrap graphics for review and approval by the NPS prior to application.

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Bus Exterior Graphics-Illustrations and Pictures for Reference Only

3.20 Underfloor

Exposed underfloor steel surfaces, after installation of welded-on bracketry, shall be undercoated or primed and undercoated.

3.21 Equipment Enclosure(s)

The exterior of all equipment enclosures shall be finished. The interior of all equipment enclosures shall be finished.

3.22 Vehicle Body Exterior

All exterior finished surfaces shall be impervious to commercial cleaning agents. Finished surfaces shall resist damage by controlled applications of commonly used graffiti-removing chemicals. Except for periodic cleaning, exterior surfaces of the bus shall be maintenance-free, permanently colored and not require refinish/repaint for the life of the vehicle. Colors and paint schemes shall be provided by the Government.

3.23 Roof

The roof shall be painted or gelcoat. If painted shall receive a minimum of a total of 4 mils of primer and topcoat as specified herein.

3.24 Exposed Piping, Conduits and Wire-ways

The contractor shall ensure that wire-ways, conduits, and piping that can be corroded shall receive a minimum of two coats of primer and two coats of an approved paint. This priming

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and painting can be accomplished either before or after installation of the item on the bus vehicle body.

3.25 Undercoating

The OEM's standard undercoating material shall be applied according to the manufacturer's instructions. Undercoating thickness shall conform to the manufacturer's recommended practices.

The composition of the materials selected for the protective treatment system shall be such that treated members may be readily cut or removed using conventional cutting methods, such as burning or sawing, without creating any fumes hazardous to employees.

All exposed conduit, piping, and bracketry shall be protected with undercoating.

3.26 Fender Skirts

Features to minimize water spray from the bus in wet conditions shall be included in wheel housing design. Any fender skirts shall be unbreakable and easily replaceable. They shall be flexible if they extend beyond the allowable body width. Wheels and tires shall be removable with the fender skirts in place.

3.27 Splash Aprons

Splash aprons, composed of composition or rubberized fabric, shall be installed with each wheel and shall extend downward. Apron widths shall be no less than tire widths and extend downward to within 6" of the road surface (static). Splash aprons shall be bolted to the bus understructure. Splash aprons and their attachments shall be inherently weaker than the structure to which they are attached. The flexible portions of the splash aprons shall not be included in the road clearance measurements. Other splash aprons shall be installed where necessary to protect bus equipment.

3.28 Windshield Wipers and Washers

The bus shall be equipped with variable speed windshield wipers complying with SAE J198. The wipers shall be electrically powered. Wiper sweep shall allow unobstructed viewing of exterior rear mirrors. If two window panes are provided, both wipers shall park along the center divider of the windshield. If a single pane is provided, the wipers shall park along the lower edge of the windshield so as not to obstruct the vision of the driver. Windshield wiper motors and mechanisms shall be easily accessible for repairs or service, mounted on mechanical fasteners, and removable as individual units from the exterior of the bus.

The windshield washer system shall deposit washing fluid on the windshield and, when used with the wipers, shall evenly and completely wet the entire wiped area.

Reservoir pumps, lines, and fittings shall be corrosion resistant, and the reservoir itself shall be translucent for easy determination of fluid level.

3.29 Service Compartments and Access Doors

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Service compartments and access doors shall have stainless steel hinges that shall be used for access to the rear compartment and all auxiliary equipment compartments. Access openings shall be sized for easy performance of tasks within the compartment, including tool-operating space. Handles shall be constructed of non-corroding material and shall be flush with, or recessed into, the body contour and shall be sized to provide an adequate grip for opening. Springs and hinges shall be stainless steel.

Access doors shall be of rugged construction and shall maintain mechanical integrity and function under normal operations throughout the service life of the bus. They shall close flush with the body surface and shall be prevented from coming loose or opening during transit service or in bus washing operations. Access doors, when opened, shall not restrict access for servicing other components or systems. The curbside and roadside rear compartment access doors shall be mounted to horizontal hinges so that they will fold up and out of the way; no vertical hinges shall be permitted.

Access doors shall be retained in the opened and closed positions with over-center gas-filled springs, without the use of latches except for electrical and radio compartments. Doors smaller than 36 square inches shall be retained in the open and closed positions by over-center springs, gas springs, or magnetic latches.

Doors shall be equipped with locking mechanisms, not keyed.

3.30 Bumpers

Bumpers shall provide impact protection for the front and rear of the bus. The bumpers shall not exceed allowable bus width. Bumper height shall be such that when one bus is parked behind another, a portion of the bumper faces will contact each other. Bumper material shall be corrosion-resistant and withstand repeated impacts of the specified loads without sustaining damage. Visible surfaces shall be black or color -coordinated with the bus exterior. These qualities shall be sustained throughout the service life of each bus.

For front bumpers, no part of each bus, including the bumper, shall be damaged as a result of a 5-mph impact of the bus at curb weight with a fixed, flat barrier perpendicular to the longitudinal centerline of the bus. The bumper shall return to its pre-impact shape. The energy absorption system of the bumper shall be independent of every power system of the bus and shall not require service or maintenance in normal operation during the service life of the bus.

For rear bumpers, no part of each bus, including the bumper, shall be damaged as a result of a 2-mph impact with a fixed, flat barrier perpendicular to the longitudinal centerline of the bus. The bumper shall return to its pre-impact shape.

The rear bumper or bumper extensions shall be shaped to preclude unauthorized riders standing on the bumper. The bumper extensions shall not hinder service and shall be faired into the body of each bus with no protrusions or sharp edges. The rear bumper shall be independent of all power systems of each bus and shall not require service or maintenance in normal operation during the service life of the bus.

Any flexible portion of the bumpers shall minimally increase the overall length of the bus vehicle.

3.31 Exterior Lighting

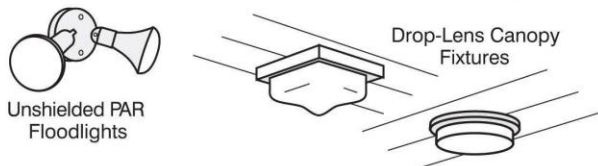
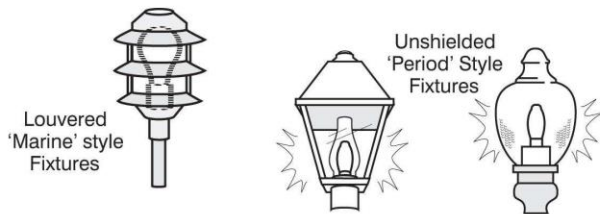
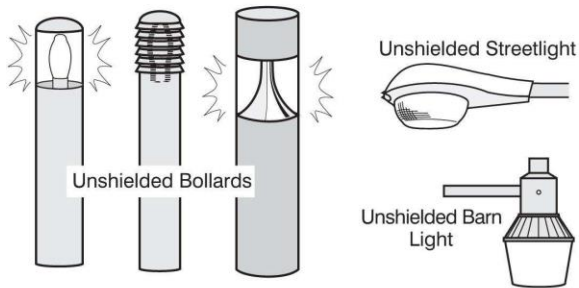
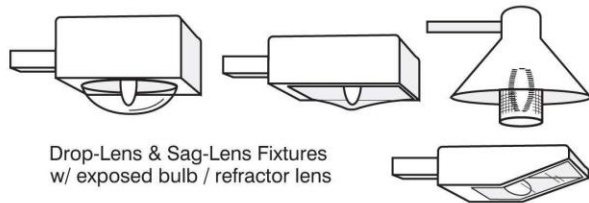
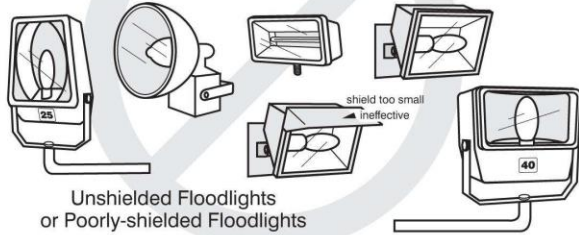
Zion National Park is a dark skies environment. For more information and guidance please refer to the following web site: <https://www.darksky.org>.

No interior and exterior lights shall be visible on the bus or the EVSE while the bus is charging between dusk and dawn. If required, the color temperature of the lights shall be 2300K, producing an amber glow. Lights fixtures shall be shielded to minimize their effects.

Examples of Acceptable / Unacceptable Lighting Fixtures

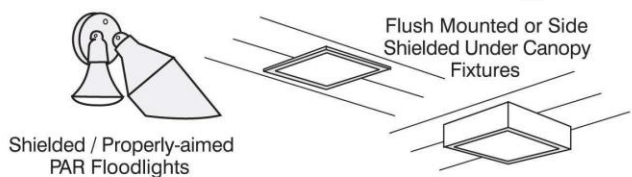
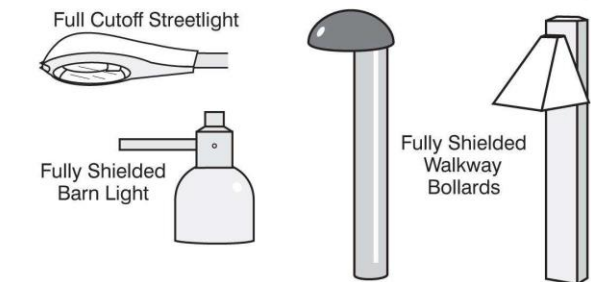
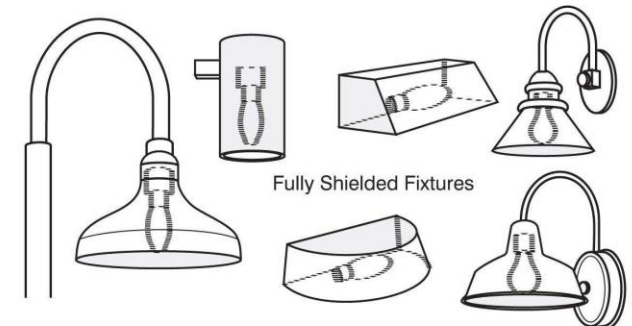
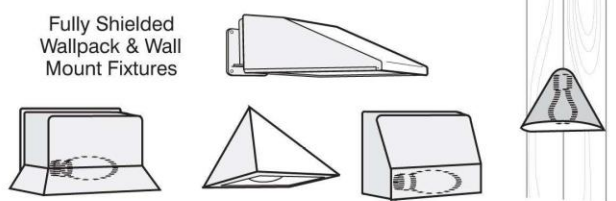
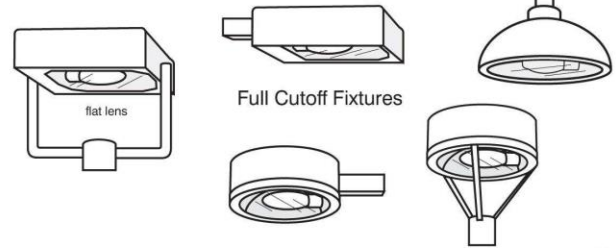
Unacceptable / Discouraged

Fixtures that produce glare and light trespass



Acceptable

Fixtures that shield the light source to minimize glare and light trespass and to facilitate better vision at night



Illustrations by Bob Crelin© 2005. Rendered for the Town of Southampton, NY. Used with permission.

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All exterior lights shall be designed to prevent entry and accumulation of moisture or dust, and each lamp shall be replaceable by a 2M mechanic within 5 minutes. Commercially available LED (Light Emitting Diode)-type lamps shall be used on all exterior lights.

Headlights shall meet applicable industry and regulatory standards, color less than 4000K and with non-LED if available.

Lights mounted on the rear compartment doors shall be protected from the impact shock of door opening and closing. Lamps, lenses, and fixtures shall be interchangeable to the extent practicable.

Hazard lamps at the rear of the bus shall be visible from behind when the rear service doors are opened. Light lenses shall be designed and located to prevent damage when running the vehicle through an automatic bus washer. Lights located on the roof and sides (directional turn signals) of the bus shall have protective shields or be of the low-profile type to protect the lens against minor impacts.

Turn-signal lights shall be provided on both sides of the bus. Visible and audible warning shall inform following vehicles or pedestrians of reverse operation. Visible reverse operation warning shall conform to SAE Standard J593. Audible reverse operation warning shall conform to SAE Recommended Practice J994 Type C or D.

3.31.1 Loading Areas

Lamps at the front and rear passenger doorways shall comply with ADA requirements and shall activate only when the doors open and shall illuminate the street surface. The lights may be positioned above or below the lower daylight opening of the windows and shall be shielded to protect passengers' eyes from glare.

The color temperature of the lights in loading areas shall be 3000K or less.

Rear cornering illumination shall be furnished on the passenger side.

3.32 Furnishings and Interior Arrangement

All furnishing design, arrangement, installation, and detail drawings shall be provided to the Government for acceptance.

3.32.1 Windshield

The windshield shall permit a driver's field of view as referenced in SAE Recommended Practice J1050. The vertically upward view shall be a minimum of 15 degrees, measured above the horizontal and excluding any shaded band. The vertically downward view shall permit detection of an object 3-1/2 feet high no more than 2 feet in front of the bus. The horizontal view shall be a minimum of 90 degrees above the line of sight. Any binocular obscuration due to a center divider may be ignored when determining the 90-degree requirement, provided that the divider does not exceed a 3-degree angle in the driver's field of view. Windshield pillars shall not exceed 10 degrees of binocular obscuration. The

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windshield shall be designed and installed to minimize external glare as well as reflections from inside the bus.

The windshield shall be easily replaceable from the bus exterior by removing zip-locks from the windshield retaining moldings. Bonded-in-place windshield shall not be used. The windshield glazing material shall have a 1/4-inch or 6-mm nominal thickness laminated safety glass conforming to the requirements of FMVSS 205 and the Recommended Practices defined in SAE J673. The glazing material shall have single density tint. The upper portion of the windshield above the driver's field of view shall have a dark, shaded band with a minimum luminous transmittance of 6 percent when tested in accordance to ASTM D-1003.

The windshield and other windows in the driver's field of vision are coated to prevent reflective glare from light sources inside the bus.

3.32.2 Driver's Side Window

The driver's side window shall open sufficiently to permit the seated driver to easily adjust the street side outside rearview mirror. The driver's window in the open position shall not block the driver's view of the exterior mirror. The movable window section shall slide rearward in tracks, channels, and seals designed in a two section sash and last the service life of the bus. The driver's side window shall not be bonded in place and shall be easily replaceable. The glazing material shall have a uniform, single density, neutral tint. Side window shall be non-whistling.

The driver's side window glazing material shall be of 1/4" laminated safety glass conforming to the requirements of the Recommended Practices defined in SAE J673 and shall meet all FMVSS requirements.

The windshield and other windows in the driver's field of vision are coated to prevent reflective glare from light sources inside the bus.

3.32.3 Passenger Windows

3.32.3.1 Materials

Side windows shall not be bonded in place, but shall be easily replaceable without disturbing adjacent windows and shall be mounted so that flexing or vibration from operation or normal road excitation is not apparent. Any aluminum and steel material will be treated to prevent corrosion. All side windows shall be easily replaceable without disturbing adjacent windows and shall be mounted so that flexing or vibration from operation or normal road excitation is not apparent.

3.32.3.2 Configuration

As a minimum, 8,000 square inches (total) of side vehicle window area (per vehicle side) shall be provided. All side passenger windows, except windows in passenger doors and those smaller than 500 square inches, shall have window panels that are openable by passengers. Openable window panels shall be equipped with latches that secure the window in the closed position. Each side window shall be a transom style window, with the top openable portion opening inward.

All side windows shall be easily replaceable without disturbing adjacent windows and shall be mounted so that flexing or vibration from operation or normal road excitation is not apparent.

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The windows shall be designed and constructed to enable a 2M mechanic to remove and replace two windows within thirty (30) minutes.

3.32.4 Emergency Exits

The bus shall be provided with adequate exits for quick passenger escape during emergency conditions. With the exception of the last three windows, two (2) windows on the street side and one (1) window on the curb side (accessible by passengers) of each bus vehicle are to be emergency egress windows and shall comply with applicable codes and requirements and the best industry practices.

Passenger window emergency exits shall open outward to provide an emergency exit. A simple red latch (or equal subject to Government review and approval) shall be provided on all emergency egress windows. This latch shall not pinch a person's fingers or hands when operating. Each emergency window location shall be labeled (preferably close to the latch).

3.32.5 Doors

3.32.5.1 General

Two (2) doorways shall be provided on the curb-side of each bus for passenger ingress and egress. The front doorway shall be forward of the front wheels. The rear doorway centerline shall be rearward to the point midway between the front door centerline and the rearmost seat back. Passenger doors and doorways shall comply with the applicable ADA requirements. Brushed stainless steel passenger assists shall be provided on all entry doors.

With the doors in any position, the driver's visibility of any mirror on each bus shall not be reduced. Doors are to be constructed to minimize vibratory noise and enhance simplified maintainability.

The front and rear doors shall be a Vapor Bus International slide-glide door and the rear door shall be a Vapor Bus International plug or slide-glide style door, furnished with Vapor Bus International's "CLASS"- Contactless Acoustic Sensing System, or equal.

3.32.5.2 Dimensions

Front and rear door width and height dimensions shall be in accordance with the successful offeror's drawing approved by the Government.

3.32.5.3 Materials

Structure of the doors, their attachments, inside and outside trim panels, and any mechanism exposed to the elements shall be durable and corrosion-resistant. Door panel construction shall be of corrosion-resistant metal or reinforced non-metallic composite materials. The doors, when fully opened, shall provide a firm support and shall not be damaged if used as an assist by passengers. Stainless steel grab handles shall be provided on the inside of all door panels.

All reasonable means shall be taken to reduce the weight of the door. The exterior of door panels shall be made of a material proven on similar buses and approved by the Government.

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Door panels, drivers, and associated drive components shall have sufficient strength and rigidity to sustain, without permanent deformation, a force of 198 pounds applied to an area of 1 square foot located 2 inches from the leading edge and centered within the door height.

3.32.5.4 Door Glazing

The upper sections of both front and rear doors shall be glazed for no less than 45 percent of the respective door opening area of each section. The front door panel glazing material shall have a nominal ¼ inch thick laminated safety glass conforming to the requirements of FMVSS 205 and the Recommended Practices defined in SAE J673, and meet or exceed the requirements of FMVSS 217.

3.32.5.5 Door Height

It shall be possible to open and close either passenger door when a bus loaded to GVWR is not knelt and parked with the tires touching an 8-inch curb on a street sloping toward the curb so that the street side wheels are 5 inches higher than the curb side wheels.

3.32.5.6 Door Sealing

The leading edges of each door panel shall form an acoustic and weather-tight seal. The edge material shall have sufficient stiffness to activate the sensitive edge control (if sensitive edge is used as a means of obstacle detection) and prevent the doors from locking when an obstacle (as defined above) is inserted between the panels.

The seal shall be readily replaceable without requiring the door panel to be removed from the vehicle. Doors shall be weather-stripped to seal against water, weather, odors, and running noise. Water shall not intrude into the bus during water spray testing. Seals shall not result in friction loading on the panels.

Doors shall be approximately flush with the bus vehicle body exterior when closed. Door panels shall be sealed against rain and bus vehicle washing. Adequate drainage from the door panels and thresholds shall be provided.

3.32.5.7 Actuators

Door actuators shall be independently adjustable, either electrically or pneumatically driven, and controlled electrically. Actuators and the door mechanism shall be concealed from passengers but shall be easily accessible for servicing.

3.32.5.8 Emergency Operations

In the event of an emergency, after actuating an unlocking device at each door, it shall be possible to open the doors manually from inside the bus using a force of no more than 25 pounds. An unlocking device shall require two distinct actions to actuate and shall be clearly marked as an emergency-only device. Precise instructions for the emergency exit shall be posted near the device, and the door emergency unlocking device shall be accessible from the doorway areas. The door interlock throttle system shall disengage electronic throttle and the door interlock brake system shall apply to stop the bus when this emergency device is actuated.

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In order to open manually, locked doors shall require a force of more than 100 pounds. Damage shall be limited to the bending of minor door linkage without resulting damage to the doors, and critical vehicle mechanisms when the locked doors are manually forced to open.

3.33 Interior Surfaces and Finishes

3.33.1 General

Interior surfaces below the lower edge of the side windows or windshield shall be shaped so that objects placed on them fall to the floor when the bus is parked on a level surface. The entire interior shall be cleanable with a hose, using a liquid soap attachment. Water and soap will not normally be sprayed directly on the instrument and switch panels. All proposed interior surface and finish material and color descriptions, as well as cleaning procedures, shall be submitted to the Government for review and acceptance. Interior materials shall meet the requirements of FMVSS 302. Alternately, interior materials shall be in compliance with FTA Docket 90-A. The bus interior shall be free of elements that limit or obstruct visibility in any direction for the driver and for passengers.

3.33.2 Interior Panels

Interior side-trim panels shall be textured anodized aluminum, melamine, or composite panels with adequate support backing. Panels shall be easily replaceable and tamper-resistant. They shall be reinforced, as necessary, to resist vandalism and other rigors of transit bus service. Individual trim panels and parts shall be interchangeable to the extent practicable. The bus interior colors shall be aesthetically pleasing and be the OEM's standard interior colors, final selection shall be reviewed and approved by the government at the VDR

3.33.2.1 Modesty Panels

Sturdy divider panels constructed of durable, unpainted, corrosion-resistant material complementing the interior trim shall be provided to act as both a physical and visual barrier for seated passengers. Modesty panels, or seats and stanchions which serve as barriers, shall be located at doorways to protect passengers on adjacent seats, and along front edge of rear upper level. Design and installation of modesty panels located in front of forward facing seats shall include a handhold/grab-handle along its top edge. These dividers shall be mounted on the sidewall and shall project toward the aisle no farther than passenger knee projection in longitudinal seats or the aisle side of the transverse seats. Modesty panels shall extend no higher than 6±1.0 inches above the lower daylight opening of the side windows and those forward of transverse seats shall extend downward to a level between 1-1/2 and 1 inch above the floor. Panels forward of longitudinal seats shall extend to below the level of the seat cushion. Dividers positioned at the doorways shall provide no less than a 2-1/2-inch clearance between the modesty panel and the opened door to protect passengers from being pinched. Modesty panels installed at doorways shall be equipped with grab rails. The modesty panel and its mounting shall withstand a static force of 250 pounds applied to a four-inch by four-inch area in the center of the panel without permanent visible deformation

3.33.2.2 Driver Barrier and Door

A barrier or bulkhead with door between the driver and the street-side front passenger seat shall be provided. The barrier shall minimize glare and reflections in the windshield directly in front of

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the barrier from interior lighting during night operation. The barrier shall extend from the floor to the ceiling and shall fit the bus side windows and wall to prevent passengers from reaching the driver or the driver's personal effects. Barrier shall be constructed of aluminum, melamine or fiberglass. It is permissible for the driver's barrier to be integrated with the driver's equipment (radio, electrical, and storage) compartment.

The barrier shall also include a center-facing door to further isolate the driver from the passengers.

3.33.2.3 Rear Bulkhead Panels

The rear bulkhead paneling shall be contoured to fit the ceiling, side walls, and seat backs so that any litter, such as a cigarette package or newspaper, will tend to fall to the floor or seating surface when the bus is on a level surface. Any air vents in this area shall be louvered to reduce airflow noise and to reduce the probability of trash or liter being thrown or drawn through the grille. If it is necessary to remove the panel to service components located on the rear bulkhead, the panel shall be hinged or shall be able to be removed and replaced by a 3M mechanic in 5 minutes. Grilles where access to or adjustment of equipment is required shall be heavy duty and designed to minimize damage.

3.33.2.4 Headliner

Ceiling panels shall be made of durable, corrosion resistant, easily cleanable material. Headlining shall be secured without loose edges and shall be supported to prevent buckling, drumming, or flexing. Where panels are in contact with metallic surfaces, headlining materials shall be treated or insulated to prevent marks due to condensation. As required to make the edges tamperproof, moldings and trim strips shall be stainless steel, aluminum, or plastic, colored to complement the ceiling material.

3.33.2.5 Fastening

Interior panels shall be attached so that there are no exposed unfinished or rough edges or rough surfaces. Panels and fasteners shall not be easily removable by passengers but shall be easily replaceable when necessary.

3.33.2.6 Insulation

Any insulation material used between the inner and outer panels shall be in accordance with this specification. Insulation properties shall be unimpaired during the service life of the bus. Any insulation material used inside the rear compartment shall not absorb or retain oils or water, shall prevent mold growth and shall be designed to prevent casual damage that may occur during maintenance operations.

The combination of inner and outer panels on the sides, roof, wheel wells and ends of the bus, and any material used between these panels, shall provide a thermal insulation sufficient to meet the interior temperature requirements. The bus body shall be thoroughly sealed so that the driver or passengers cannot feel drafts during normal operations with the passenger doors closed.

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3.33.2.7 Front End

The entire front end of the bus shall be sealed to prevent debris accumulation behind the dash and to prevent the driver's feet from kicking or fouling wiring and other equipment. The front end shall be free of hazardous protrusions. Paneling across the front of the bus and any trim around the driver's compartment shall be formed metal, ABS plastic or reinforced fiberglass.

Plastic dash panels shall be reinforced, as necessary, resistant to age discoloration and cracking, vandal-resistant, and replaceable. All colored, painted, and plated parts forward of the driver's barrier shall be finished with a smooth, dull matte surface that matches or coordinates with the bus interior.

3.33.2.8 Access Panels and Doors

Access for maintenance and replacement of equipment shall be provided by panels and doors that appear to be an integral part of the interior. Removal of fixtures and equipment that are unrelated to the repair task to gain access should be minimized. Where possible, access doors shall be hinged with props to hold the doors out of the mechanic's way. All door hinges shall be stainless steel.

3.34 Floor Covering

The floor covering shall have a non-skid walking surface that remains effective in all weather conditions. The floor covering, as well as transitions of flooring material to the main floor and to the entrance and exit area, shall be smooth and present no tripping hazards. Seams shall be sealed/welded per manufacturer's specifications. The rubber floor covering shall be R.C.A. model TransitFlor, or equal. The floor covering layout, tread patterns, and colors shall be selected and approved at the VDR.

The area of the front ramp platform as well as the floor area under and around the ramp in the vestibule area may be sprayed-on polyurethane, non-skid surface. The step edge shall be yellow. Any areas on the floor, which are not intended for standees, such as areas "swept" during passenger door operation, shall be clearly and permanently marked. The floor in the driver's compartment shall be easily cleaned and shall be arranged to minimize debris accumulation.

Prior to applying floor covering, all voids, fastener heads, and cracks between floor panels shall be filled, and the floor made smooth.

Seams shall be tightly bonded to the floor panel and have no perceivable variations in heights. At all door openings, the floor covering shall connect properly and form a positively clamped, watertight seal with mating moldings/threshold plates.

Standee lines shall be marked in yellow at both the front vestibule area and rear door area of the bus. The standee line at the front door shall be a minimum 2 inches wide and at the rear door area shall be a minimum of 5 inches wide.

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3.35 Passenger Interior Lighting

Interior lighting shall be located overhead and under passenger seating.

All interior lighting shall be LED configuration, with a color temperature of 2300K, producing an amber glow.

When in the RUN and NITE/RUN mode, all overhead lighting shall automatically extinguish or dim when the front is in the closed position and all interior modules shall light when the front or rear doors are opened.

In addition to the automatic control specified above, a separate manual switch shall allow the driver to control the overhead lights- on/off. A similar manual switch shall allow the driver to control the under seat lighting- on/off.

The interior lights shall be dimmable.

Interior Entry Area Lighting

The bus shall be equipped with lights to illuminate the doorways and adjacent platform for the purpose of safe boarding and disembarking of passengers.

3.36 Passenger Seats

3.36.1 Passenger Seats General

There shall be a minimum total of 34 passenger seats, including fold-down seats to accommodate wheelchairs. The seating configuration shall be perimeter type and shall not exceed seven (7) forward facing seats. While it is anticipated that these buses will not require passenger seat belts, if after the order is placed, the manufacturer determines that FMVSS compliance requires seat belts, they will be added at additional cost negotiated between the Government and the contractor.

The seats shall be, American Seating Insight Prime +, or equal, thermoplastic seats with stainless steel frame, two-tone inserts, grab rail, drain hole, (meeting FMVSS 302 and FTA Docket 90-A). The color of the seats shall be pewter grey. The successful offeror shall furnish a seating layout drawing which will be approved by the Government.

The seating layout, make/model, and color shall be reviewed and approved by the government.

As applicable, hip-to knee room measured from the front of one seat back horizontally across the highest part of the seat to the seat or panel immediately in front shall be no less than 26 inches. At all seating positions in paired transverse seats immediately behind other seating positions, hip-to-knee room shall be no less than 26 inches. Seat height, except for rear bench seat, shall be 17 inches. The rear bench seating height shall not exceed 21 inches.

Transverse mounted seats shall, wherever possible, be fully cantilever mounted.

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3.37 Wheelchair Accommodation

Two (2) forward-facing wheelchair securement locations and a total of two securements, as close to the wheelchair loading system as practical, shall be provided for each bus. Each wheelchair accommodation shall provide parking space and a secure tie down (compliant with ADA requirements) for two passengers in wheelchairs. Q-Straint Incorporated, or equal, self-tensioning, continuous adjustment wheelchair tie-down system with combination retractable lap and shoulder belt, shall be provided.

3.37.1 Maneuvering Room

Maneuvering room inside the bus shall accommodate ease of travel for a passenger in a wheelchair from the loading device through the bus to the designated parking area and back out per ADA requirements.

Additional equipment, including passenger restraint seat belts, shoulder harnesses and wheelchair securement devices shall be provided for each wheelchair passenger. Passenger restrain seat belts shall be provided to accommodate passengers in electrically powered wheelchairs. All belt assemblies shall stow up and out of the way when not in use.

3.37.2 Wheelchair Securing Devices

Two (2) wheelchair securing devices shall be provided. The installation of wheelchair securement device shall not require excessive bending or twisting by the driver. The jump seat release mechanism shall be easily operable.. The latching mechanisms for wheelchair wheels shall not interfere with battery-operated wheelchairs.

3.37.3 Seat Belts

Built-in passenger restraint seat belts shall be provided in each wheelchair parking area. Seatbelts shall be easily accessible for wheelchair users. On the curbside securement area a shoulder belt should be attached to the sidewall of the bus. A belt-type securement system and shoulder strap seat belt shall be included. The latching mechanism and retracted belts shall be readily visible when seats are folded down. Wheelchair area accommodations shall comply with the latest ADA laws and federal safety requirements.

3.38 Passenger Assists

Stanchions and handrails shall be provided to aid passengers when boarding, moving through the vehicles, and standing while onboard the vehicles.

Stanchions shall not be placed where they will obstruct the passenger flow through the bus vehicle doors. Stanchions shall be designed to adequately withstand the loads that may be applied to them (up to 300 pounds at midpoint in any direction) without permanent deformation. Stanchions and handrails shall be brushed stainless steel or approved equal. Stanchions and handrails shall comply with ADA regulations and shall be between 1.25 inches and 1.5 inches in diameter unless otherwise approved.

Brushed stainless steel passenger assists shall be provided in the articulated joint section to aid in the transition between the front and rear sections of the bus.

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A minimum of 24 overhead grab straps, color grey, shall be provided.

3.39 Wheelchair Loading System

3.39.1 General

A loading system shall accommodate passengers on wheelchairs or using crutches, canes, walkers, or persons with difficulty using steps for ingress and egress from or to the street level or curb quickly, safely, and comfortably. The wheelchair loading system shall conform to all applicable ADA requirements.

3.39.2 Wheelchair Ramp

A front door located, automatically controlled, ramp system, ADA compliant, meeting requirements defined in 49 CFR Part 38, shall provide ingress and egress quickly, safely, and comfortably, both in forward and rearward directions, for a passenger in a wheelchair from a level street or curb. The ramp shall be of a simple hinged, flip-out type design. The deployed ramp shall be 30 inches wide (minimum) and 44 inches long (minimum).

When the system is not in use, the passageway shall appear normal. In the stored position of the ramp, no tripping hazards shall be presented and any resulting gaps shall be minimized. The controls shall be simple to operate with no complex phasing operations required, and the loading system operation shall be under the surveillance and complete control of the driver. The bus shall be prevented from moving during the loading or unloading cycle by a throttle and brake interlock system. The wheelchair loading system shall not present a hazard, nor inconvenience any passenger. The loading system shall be inhibited from retracting or folding when a passenger is on the ramp/platform. A passenger departing or boarding via the ramp shall be able to easily obtain support by grasping the passenger assist located on the doors or other assists provided for this purpose. The platform shall be designed to protect the ramp from damage and persons on the sidewalk from injury during the extension/retraction or lowering/raising phases of operation. The loading platform shall be covered with a replaceable or renewable, nonskid material and shall be fitted with devices to prevent the wheelchair from rolling off the sides during loading or unloading. A manual override system shall permit unloading a wheelchair and storing the device in the event of a primary power failure.

3.40 Driver Area

The objective of the design of the driver's area is to provide an environment for the driver to operate the bus safely and efficiently for long periods of time without injury and with minimal fatigue. The driver's area shall therefore be designed to minimize driver glare. The use of polished metal and light-colored surfaces within and adjacent to the driver's area shall be avoided. Areas that are visible from the outside of the bus in the vicinity of the dash panel and cowl shall be configured to preclude use as storage of items.

The driver's area shall comply with the following SAE recommended practices, including:

- SAE J287, Driver Hand Control Reach;
- SAE J941, Motor Vehicle Driver Eye Range;
- SAE J1050, Driver's Field of View;

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- SAE J1052, Motor Vehicle Driver and Passenger Head Position;
- SAE J1516, Accommodation Tool Reference Point;
- SAE J1522, Truck Driver Stomach Position.

3.40.1 Driver Controls/Instrumentation

All switches and controls necessary for the safe and efficient operation of the bus vehicle shall be conveniently located in the driver's area, grouped according to function. Controls shall be located so that boarding passengers may not easily tamper with control settings. Similarly, all switches and controls shall not pose a hazard to the Driver. Guards shall be provided on switches as necessary to prevent inadvertent activation. The nomenclature of each indicator shall be readable when either energized or de-energized. All controls/instruments shall be placed and protected to prevent damage by wind or rain when any bus vehicle windows/doors are open, whether the bus vehicle is moving or is stationary.

3.40.2 Controls

Bus startup and shutdown procedure shall be intuitive and accomplished within 15 minutes (including building air pressure for pneumatic systems). Status indicators/gauges, including time remaining, before vehicle is drivable or completely shut down shall be provided.

As located in ergonomically convenient locations for the driver, the following identified switches and lights shall be visibly accessible and include (at a minimum):

- Master run switch;
- Start button or switch;
- Kneel switch;
- Interior lighting switch;
- Instrument panel lighting intensity control (solid state);
- Driver's area light switch;
- Hazard warning switch;
- Climate controls;
- Defroster control;
- Driver's heater control;
- Parking emergency brake control (brake activation indicated);

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- Front door 'dump' valve;
- PA system controls;
- High beam;
- Exit door open or unlocked;
- Horn button in steering wheel hub;
- Headlight dimmer switch (foot controlled);
- Wheelchair ramp switches and ramp control switch;
- Driver's fan;
- Motor 'Stop' override;
- Front dome light.

Covert alarm switch

Cup-holder

Exterior bell with momentary switch, approach alert for bicyclists

These controls shall be identifiable by shape, touch, and permanent markings. All switches shall be heavy-duty push button, rocker, rotary (spring return to center) or toggle switches. Requests for other types of switches or controls shall be submitted to the Government for review and approval. Door control, kneel control, windshield wiper/washer controls, and run switch controls shall be the most convenient for the driver to access. The hazard-warning switch shall be longer or physically different thereby allowing the driver to quickly find its location and activate it merely by feeling for the switch. Doors shall be operated by a single control, conveniently located and operable in a horizontal plane by the driver's left hand. The setting of this control shall be easily determined by position and touch. Turn signal controls shall be floor-mounted, foot-controlled, water resistant, heavy-duty, momentary contact switches.

Switches, controls, and instruments shall be dust- and water-resistant consistent with the specified bus interior cleaning practice.

3.40.3 Instruments

The speedometer, state of charge indicator, certain indicator lights, and air pressure gauge(s) with two needles shall be a minimum of two (2) inches in diameter and shall be located on the front cowl immediately ahead of the steering wheel. Illumination of the instruments shall be simultaneous with the marker lamps. Glare or reflection in the windshield, side window, or front door windows from the instruments, indicators, or other controls shall be minimized. Instruments

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and indicators shall be easily readable in direct sunlight. The use of backlit instrumentation and gauges shall be maximized.

The instrument panel shall include an electric analog speedometer or digital display with a maximum speed over 65 mph and calibrated in maximum increments of 5 mph.

The instrument panel shall also include an air pressure gauge with indicators for primary and secondary air tanks, voltmeters for 24volt systems, and any other relevant gauges. The instrument panel and wiring shall be easily accessible for service from the driver's seat or top of the panel. Wiring shall have sufficient length and be routed to permit service without stretching or chafing the wires.

3.40.4 Indicators/On-Board Diagnostics

Critical systems or components shall be monitored by a built-in diagnostic system with visible and audible indicators. The diagnostic indicator lamp panels shall be located in clear sight of the driver. The intensity of indicator lamps shall permit easy determination of on/off status in bright sunlight and shall not cause a distraction or visibility problem at night. All indicators shall have a method of momentarily testing the operation of the lamp. The successful offeror shall submit the design, installation, and operating details of the proposed diagnostic system indicator and on-board diagnostics display(s) to the Government for review and acceptance.

Whenever possible, sensors shall be of the closed-circuit type so that failure of the circuit or sensor shall activate the malfunction indicator. Audible alarms shall be loud enough for the driver to be aware of its operation and be inclined to discontinue operation of the bus. Malfunction and other indicators listed in Table 2-Onboard Diagnostic Indicators shall be provided for the bus:

Table 2: Onboard Diagnostic Indicators

Visual Indicator	Audible Alarm	Condition or Malfunction
ABS	None	ABS System Malfunction
Hazard Warning	Audible Clicks	Warning signal to other drivers (may be common with turn indicators)
High Headlamp	None	High headlamp is on
Kneeling	Buzzer	Kneeling system activated
Low Charge	None	Low charge conditions
Low Air	Buzzer	Insufficient air pressure in either primary or secondary reservoirs
Mobility Aid Passenger Exit Signal	None	Mobility aid passengers want to disembark
Parking Brake Applied	None	Parking brake is applied
Service Brake Applied	None	Service brake is applied (may be common with parking brake indicator)
Rear Doors Open	Buzzer	Rear doors are opened

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Reserved		
Right and Left Turn	Audible Clicks	Indication of left or right turn
Wheelchair Ramp	Beeper	Wheelchair ramp is not stowed and disabled

3.40.5 Door Controls

Controls for the front and rear doors shall be a single, 4 or 5-position master door switch, conveniently located and operable in a horizontal plane by the driver's left hand. The setting of this control shall be easily determined by position and touch. Other concepts may be proposed for Government consideration. The master door switch shall be a single, 4 or 5-position, control with the following settings provided:

<u>Switch/Lever Position</u>	<u>Front Door</u>	<u>Rear Door</u>
1 st Position Rearward	Closed	Opened
Center Position	Closed	Closed - Locked
1st Position Forward	Opened	Closed – Locked
2nd Position Forward or Rearward	Opened	Opened

3.40.5.1 Door Operations

The front bus vehicle doors are to be 'driver open' and 'driver closed' commanded. Power for opening and closing the front doors shall be done either pneumatically or electrically. The door shafts shall be designed to mitigate operational problems (such as 'jumping' out of their bearings). The rear door is also to be 'driver open' and 'driver closed' commanded. Opening the rear door shall be conducted either pneumatically or electrically with a single cylinder, closed with a spring or air cylinder, and locked with an electro-mechanical lock.

No more than a 10 pound force shall be imposed on a 1 square inch area of any passenger struck by a closing rear door. Sensitive edges are to be verified with a 1.5-inch obstruction that simulates a small child's arm. A maximum force of 25 pounds shall be required for a passenger to free himself after having either door close upon him/her. The opening and closing of all doors should take no more than 4 seconds.

Door mechanisms shall be concealed from passengers but shall be easily accessible for servicing. All elements of the door and actuator systems shall operate trouble-free for the life of the vehicle.

For air-powered doors, a valve, located at the top of the left side panel and accessible through the driver's window, shall shut off the air power to, and dump the air power from, the front door mechanism to permit manual operation of the front door with the bus shut down. Operation and location of the door dump valve shall be accepted by the Government during preliminary design review activities.

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3.40.5.2 Interlock

An accelerator interlock system shall be furnished that disengages the electronic accelerator and a braking interlock shall engage the rear axle spring brake system in accordance with the following door and door control lever positions:

<u>Door Lever Position</u>	<u>Front Door</u>	<u>Rear Door</u>	<u>Interlocks</u>
1 st Position Rearward	Closed	Opened	Actuated
Center Position	Closed	Closed – Locked	Not Actuated
1st Position Forward	Opened	Closed – Locked	Not Actuated
2nd Position Forward or Rearward	Opened	Opened	Actuated

3.40.5.3 Interlock Override Switch

An interlock override switch shall be integrated to release and deactivate all interlocks. The override switch shall be such as to prevent unintentional activation. Access to the interlock override switch shall be accepted by the Government during preliminary design review activities.

3.40.6 Steering Wheel and Horn Button

The steering wheel shall last the life of the bus and shall be constructed of a hard, smooth material impervious to cleaning fluids, and body acids. The steering wheel shall be of an effective diameter and shall be shaped for firm grip with comfort for long periods of time. The steering wheel spokes or rim shall not obstruct the driver's vision of the instruments on the panel when the steering wheel is in the straight-ahead position. The steering column shall be capable of horizontal and vertical adjustment from the driver seat. Clearance requirements shall be met at all positions with adjustment ranges.

Dual electric horns shall be provided. They shall be mounted to prevent the entry of water and dirt into horn trumpets, the horns shall sound high and low notes that are clearly audible over 75 dBA noise at a distance of 300 feet. The horn button shall be located in the steering wheel hub and shall be protected from debris accumulation.

3.40.7 Accelerator and Brake Pedals

Accelerator and brake pedals shall be designed for comfortable ankle motion and shall meet the requirements of SAE J1516. Foot surfaces of the pedals shall be faced with wear-resistant, non-skid, replaceable material. Force to activate the brake pedal control shall be an essentially linear function of the bus deceleration rate and shall not exceed 70 pounds at 7 inches above the heel point of the pedal to achieve maximum braking. The heel point is the location of the driver's heel when foot is rested flat on the pedal and the heel is touching the floor or heel pad of the pedal.

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3.40.8 Master Run Switch

Controls for electric drive operation shall be closely grouped within the driver's area. These controls include a separate master run switch and start switch or button. The master run switch shall be a four-position rotary switch (or equivalent) conveniently located to the driver's left with the following functions:

STOP MOTOR - All electrical systems shall be turned off by the PLC after a 30 minute delay, except hazard lights, radio, systems which shall remain energized

NIGHT PARK - All electrical systems off, and power to destination signs, interior lights, radio and marker lights.

DAY RUN - All electrical systems on, except the headlights, parking lights and marker lights.

NIGHT RUN - All electrical systems on.

3.40.9 Turn Signal

Turn signal controls shall be foot controlled, waterproof, heavy-duty momentary contact switches, and floor-mounted on a platform inclined to minimize confusion among the left, right, and high-beam switches.

3.40.10 Driver's Area Lighting

The driver's area shall have a light to provide general illumination and it shall illuminate the half of the steering wheel nearest the driver to a level of 10-15 foot candles. This light shall be driver controlled by a toggle switch located on the driver's control panel.

3.41 Sun Shades

A retractable, adjustable, mesh shades, shall be provided for use on the driver's side of the windshield and the driver's side window. The shades shall be shaped to minimize light leakage between the shade and windshield pillars. They shall store out of the way and shall not obstruct airflow from the climate control system or interfere with other equipment such as the destination sign control. Adjustments shall be made easily by hand with positive locking and releasing devices and shall not be subject to damage by over tightening. Sun shade construction and materials shall be strong enough to resist breakage during adjustments.

3.42 Driver's Seat and Driver Protection Door

The driver's seat shall be high back with head rest and include lumbar support. The seat back shall be cloth. The driver seat shall be Recaro Ergo M Series, or equal.

The belt assembly should be an auto-locking retractor (ALR). All seat belts should be stored in automatic retractors. The belts shall be mounted to the seat frame so that the driver may adjust the seat without resetting the seat belt. The seat and seat belt assemblies as installed in the bus shall withstand static horizontal forces as required in FMVSS 207 and 210.

Color selection and final configuration shall be reviewed and approved by the government.

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A Wabco Vapor V Shield, ArrowGuard, or equal, driver protection door shall be provided. The door shall include integrated cooling fans, an electric powered or sliding window, laminated safety glass, and a latch.

3.43 Driver's Vent and Defroster

The bus vehicle interior climate control system shall deliver at least 500 cubic feet per minute of air to the driver's area when operating in the ventilating mode. Adjustable nozzles shall permit variable distribution or shut down of the air flow. Air flow in the heating mode shall be reduced proportionally to the reduction of air flow into the passenger area.

The windshield defroster unit shall meet the requirements of SAE Recommended Practice J382 and shall have the capability of diverting heated air to the driver's feet and legs. The defroster or interior climate control system shall maintain visibility through the driver's side window. A draining tube shall be installed if the defroster or driver's heater is to be in an enclosed box.

3.44 Auxiliary Windshield De-Mister Fans

Two (2) auxiliary, two-speed fans shall be at the front windshield area. If mounted on the dash, fan bases shall be mounted to a tapping plate that is installed to reinforce the dashboard. These plates shall be attached by welding or with machine screws. No rivets are allowed. Each fan shall be equipped with a "HIGH-OFF-LOW" toggle switch located convenient to the driver to selectively control the operation of the fan.

3.45 Interior Mirrors

Mirrors shall be provided for the driver to observe passengers throughout the bus without leaving his/her seat and without shoulder movement. With a full standee-load, including standees in the vestibule, the driver shall be able to observe passengers in the front and rear door area, anywhere in the aisle, and in the rear seats. Interior mirrors shall not be in the line of sight to the right outside mirror. Mountings shall be sturdy to resist flexing, vibration, and vandalism.

Interior observation shall be accomplished by a swivel-mounted mirror no less than 4 inches by 16 inches attached above and to the right of the driver's head. The rear exit shall be observed by the use of two mirrors. One shall be a 6-inch round mirror installed at the upper right corner near the windshield header. The second shall be a 12-inch convex mirror installed at the rear exit. Mirrors mounted near the entrance or exit shall not be an obstruction to passengers and shall provide maximum visibility to the driver.

3.46 Exterior Mirrors

The bus shall be equipped with a corrosion-resistant exterior rearview mirror on each side of the bus. Mirrors shall permit the driver to view the roadway along both sides of the bus, including the rear wheels. Each mirror shall be heated and shall feature motorized adjustment (being remote controlled in an ergonomically efficient manner relative to a seated driver). Mirrors shall be firmly attached to the bus to prevent vibration and loss of adjustment, but not so firmly attached that a bus or its structure is damaged if the mirror is struck, and shall retract or fold sufficiently to allow bus washing operations. Mounting arms shall not protrude beyond the outside mirror edge.

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3.47 PA/Radio + Passenger Counter Provisions+ Annunciator

3.47.1 Public Address System

A public address system that complies with the ADA requirements of 49 CFR, Part 38.35 and enables the driver to address passengers either inside or outside the bus shall be provided. The system shall be comprised of one (1) audio amplifier, one (1) minimum 30inch remote mounted microphone with on/off switch, loudspeakers as necessary, one (1) Inside/Outside selector switch, and cable, connectors, and hardware as necessary to install a working system as described herein.

Inside speakers shall broadcast, in a clear tone, announcements that are clearly perceived from all seat positions at approximately the same volume level. The loudspeakers shall be located inside the bus and distributed in a fashion so as to provide a nominal level at each seat with no more than 3 dbA variation seat-to-seat. The loudspeakers selected for use inside the bus shall meet the following minimum requirements:

A speaker shall be provided so announcements can be clearly heard by passengers standing outside each bus near the front door. One or more loudspeakers shall be provided and installed where necessary to provide announcements to passengers outside the bus and within 25' of the bus doors. These loudspeakers shall be impervious to all types of weather as well as bus wash operations. Loudspeakers shall be selected and installed so that they do not obstruct the view of any seated passenger or the driver. The loudspeakers shall not be selected, located, or installed in such manner that they may cause harm or injury to any person walking, standing, or sitting near any loudspeaker in the course of normal use and operation.

The public address system shall be self-muting when not in use and powered from a source which does not have adverse electrical noise. The contractor shall employ measures as necessary to defeat any noise intrusion into the sound system such as pops, buzzes, clicks, hum, hiss, and ignition noise as attributed but not limited to power source, alternator, generator, or lighting noise without alteration to the sound system performance.

The microphone shall be vandal resistant and mounted on a heavy-duty flexible gooseneck, minimum 24 inches long, and shall allow the driver to comfortably speak into it without holding it. The gooseneck mounting flange shall be located at the contractor's standard mounting location.

A jack and a hand held microphone with push-to-talk-button shall be provided on the forward curb side exterior of the bus.

3.47.2 Two Way Radio Provisions

The contractor shall provide and install 12VDC 20A ampere power provisions (B+ & ground), a ground plane on the bus roof, two (2) NMO mounts installed on the roof ground plane with minimum 18 inches separation, antenna coaxial cables terminated with mini UHF male crimp connectors both ends routed from NMO mounts to driver area, coaxial cables shall have 24 inch service loops on both ends, mounting provision for NPS base radio (Motorola XPR4550) , mounting provisions NPS Motorola radio telephone style handset (Motorola Model 111 Handset Kit HMN4098A , and conduit with spare pull wire.

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The customer shall provide and install, after acceptance and delivery, the radio, handset, and antennas of their choosing.

3.47.3 Passenger Counter Provisions

The contractor shall provide and install power provisions above each door, mounting provisions, a suitable cradle for the NPS tablet, and a conduit with pull wire. The tablet mounting provisions shall be located so the tablet is viewable/touchable from the driver position.

The customer shall provide and install, after acceptance and delivery the tablet of their choosing

NPS Tablet	MFG TBD	Model TBD	QTY
Samsung 10.4 inch Tab A's			1

3.47.4 Annunciator Provisions

An annunciator system shall be provided. The system shall be a Luminator Cab Control Unit with ICU 403 Passenger Information System, or equal.

The system shall be compatible with destination signs, side and rear displays. The controller shall support over the air wireless updates.

The contractor shall furnish, install, and integrate all hardware/software/components to provide a turn-key passenger annunciator solution.

The Government reserves the right to review and approve the complete solution at the design review meeting.

3.48 Storage Locker

The contractor shall furnish and install a minimum of one driver's storage locker. The locker(s) shall be furnished with finger operated latches-no keyed locks.

3.49 Driver's Platform

The driver's platform shall be finished with no sharp edges and shall not interfere with or impede wheel chairs or other mobility aids. The floor in the driver's area shall be easily cleaned and shall be arranged to prevent debris accumulation.

3.50 Front Curbside Wheelwell Cover and Package Rack

The front curbside wheelwell cover shall be a "square top" wheelwell cover furnished with a large stainless steel package rack.

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3.51 Heating, Ventilation, and Air Conditioning

3.51.1 Configuration

The HVAC unit shall be a rear or roof mounted. The system shall be furnished with the following:

- Electric compressor
- Brushless evaporator and condenser fan motors
- Chlorine free, non-ozone depleting refrigerant
- Climate control system

Manually controlled shutoff valves in the refrigerant lines located on either side of all circulating/booster pumps shall be implemented to facilitate ease of service. Self-sealing couplings utilizing O-rings seals, or shut off valves, shall be used to break and seal the refrigerant lines during major components removal.

The condenser shall be located to efficiently transfer heat to the atmosphere without ingesting air warmed by other components and discharging air to other components. The condenser shall also be obstruction free from wheel splash, road dirt, or debris.

All HVAC components located within 6 inches of floor level shall be constructed to resist corrosion and damage.

3.51.2 Performance and Capacity

With the bus operating at its design operating profile and door opening/closing cycle, and at 150 % of seated load, the HVAC system shall maintain an average passenger compartment temperature between 65 degrees and 80 degrees Fahrenheit, and a relative humidity of 50% or less.

When the bus is operated in outside ambient temperatures of 95 degrees to 115 degrees Fahrenheit, the interior temperature of the bus shall be permitted to rise 1 degree for each degree of exterior temperature in excess of 95 degrees Fahrenheit.

At an outside ambient temperature range of -10° to +110° F, the interior temperature of the bus shall not fall below 55° Fahrenheit, while operating on the Design Operating Profile. If required to meet these requirements, a supplementary propane (LPG) or diesel fuel fired air heater system shall be furnished. The supplemental system shall include the heater, fuel tank, piping, ducting, and controls.

The air conditioning portion of the HVAC system shall be capable of reducing the passenger compartment temperature from 110° to 90° F in less than 20 minutes after start-up. During the cool-down period the refrigerant pressure shall not exceed safe high-side pressures and the condenser discharge air temperature, measured 6 inches from the surface of the coil, shall be less than 45° F above the condenser inlet air temperature. The appropriate solar load as recommended in the APTA "Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning System," representing 4 P.M. on August 21, shall be used. There shall be no passengers on board, and the doors and windows shall be closed.

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The pull up requirements for the heating system shall be in accordance with Section 10 of APTA's "Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning." With an ambient temperature of -10° F, and the bus cold soaked at that temperature, the bus heating system shall warm the center passenger compartment to an average temperature of $70^{\circ} \pm 2^{\circ}$ F within 70 minutes. An auxiliary propane fuel fired air heating system (heater, controls, and fuel tank) may be installed on the bus and utilized if needed to meet this performance requirement.

Testing shall be conducted in accordance to the APTA Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning System. Temperature measurements shall be made in accordance to this document with the following modifications:

The three primary locations used for temperature probes are (1) 6 inches aft of front wheel housing, (2) centered between the two axles and (3) 6 inches aft of rear wheel housing. At each primary location, the nine (9) temperature sensing devices shall be (A) 72 inches above floor level, (B) 6 inches above top surface of seat cushion and (C) 6 inches above floor.

The recommended locations of temperature probes are only guidelines and may require slight modifications to address actual bus design. Care shall be taken to avoid placement of sensing devices in immediate path of air duct outlet. In general, the locations are intended to accurately represent the interior passenger area.

3.51.3 Controls

The HVAC system, excluding driver's heater and defroster, shall be centrally controlled with an advanced electronic/diagnostic control system having provisions for extracting data via a portable testing unit.

The temperature control set-point for the system shall be 70 degrees Fahrenheit. The entire HVAC system shall be operated fully automatically to control the interior passenger compartment average temperature to within 2 degrees Fahrenheit of the aforementioned temperature control set-point. In the event of compressor or condenser fan failures, the control system shall automatically switch to ventilating mode.

Temperatures measured from a height of 6 inches below the ceiling shall be within ± 5 degrees Fahrenheit of the average temperature at the top of the seat surfaces. Temperatures measured more than 3 inches above the floor shall be within ± 10 degrees Fahrenheit of the average temperature at the top of the seat surfaces. The interior temperatures measured from front to rear of the bus shall not vary more than ± 5 degrees Fahrenheit from the average when in heating mode.

Vents shall be provided to supply conditioned air into the driver's area. The vents and fans leading to the driver's area shall be controlled by the driver. Further, the driver shall have full control over the driver's heater and defroster, and the temperature in his/her area shall be adjustable through air distribution and fans.

3.51.4 Air Flow

Ventilation shall be accomplished by the blower fans that are integral within the HVAC system.

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The cooling mode of the HVAC control system shall recirculate air within the bus at or near the ceiling height at a minimum rate of 25 cubic feet per minute (cfm) per passenger based upon the standard configuration bus carrying a number of passengers equal to 150 percent of the seated/standing load. Air flow shall be evenly distributed throughout the bus with air velocity not exceeding 100 feet per minute on any passenger. The ventilating mode shall provide air at a minimum flow rate of 20 cfm per passenger.

Air flow may be reduced to 15 cfm per passenger (150 percent of seated load) when operating in the heating mode. The fans shall not activate until the heating element has warmed sufficiently to assure 70 degrees Fahrenheit air outlet temperature. The heating air outlet temperature shall not exceed 120 degrees Fahrenheit under any normal operating conditions. Outside air flow may be cut off during initial warm-up, provided no manual manipulation is required.

In addition, the HVAC system shall deliver at least 100 cfm of air to the driver's area when operating in the ventilating and cooling modes. Adjustable nozzles shall permit variable distribution or shutdown of the airflow. Airflow in the heating mode shall be reduced proportionally to the reduction of airflow into the passenger area.

The windshield defroster unit shall meet the requirements of SAE Standard J382 and be able to divert heated air to the driver's feet and legs. The system shall also maintain visibility through the driver's side window.

3.51.5 Emergency Escape Hatch

A minimum of one 1)combination roof ventilator/escape hatch shall be provided in the roof of each bus vehicle. The ventilator/hatch shall comply with the requirements of FMVSS 217. The ventilator/escape hatch shall cover an opening area no less than 425 square inches and shall be capable of being positioned as a scoop with either the leading or trailing edge open or with all four edges raised simultaneously.

3.51.6 Air Intakes

Outside openings for air intake shall be in a location designed to ensure cleanliness of air entering the HVAC system, particularly with respect to exhaust emissions from the bus, adjacent traffic, and airborne dust generated by the rear wheels. All intake openings shall be baffled to prevent entry of water.

In addition, outside air shall be filtered before discharge into the passenger compartment. The filter shall meet ASHRAE requirement for 5 percent or better atmospheric dust spot efficiency and a minimum dust holding capacity of 120 grams per 1000-cfm cell. More efficient air filtration may be provided to maintain efficient heater and/or evaporator operation. Air filters shall be easily removed for service. Moisture drains from air intake openings shall be located to prevent clogging from road dirt.

3.51.7 Entrance/Exit Area Heating

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Heat shall be supplied to the entrance and exit floor areas of the proposed bus vehicle to prevent accumulation of snow, ice, or slush with bus operating under design operating profile and corresponding door opening cycle.

3.52 Signage

All electrical materials, components and installation methods used in the fabrication of the entire communication/signage system, including wire, cable, printed circuit boards, antennas, and circuit protection devices shall conform to this Section.

3.52.1 Destination Signs

3.52.2 General

A programmable destination sign system shall be furnished on the front and the right side near the front door of the vehicle. All signs shall be controlled via a single human-machine interface (HMI). In the absence of a single mobile data terminal (MDT), the HMI shall be conveniently located for the bus driver within reach of the seated driver.

The sign located near the front door shall not block the driver's critical horizontal line of sight. Display areas of destination signs shall be clearly visible in direct sunlight and/or at night. Signs shall be installed to allow replacement by a 3M mechanic within 30 minutes. Parts shall be commercially available. External passenger information signs shall meet all applicable Federal ADA requirements.

The destination sign compartments shall be designed to meet the following minimum requirements:

- Prevent condensation and entry of moisture and dirt.

- Prevent fogging of both compartment window and glazing on unit itself.

- Access shall be provided to allow cleaning of inside compartment window and unit glazing.

- Front window shall have an exterior display area of no less than 8.5" height x 56" width.

3.52.3 Other Numbering and Signage

Signs shall be durable and fade, chip, and peel resistant - they may be painted signs, decals, or pressure-sensitive applied. All decals shall be sealed with clear waterproof sealant around all exposed edges (as required). Signs shall be provided in compliance with applicable ADA requirements as defined in 49 CFR subpart B, 38.27.

3.53 Passenger Information Holder

Provisions shall be made at the rear of the driver's barrier for a frame to retain information posted by Government, such as routes and schedules. The retainers may be concave and shall support the media without adhesives.

In addition, provisions shall be furnished for interior overhead bus cards, which permit the Government to install ad-style displays running up both sides of the interior of the bus. These display holders shall support the display media without adhesive.

3.54 Low Voltage Electrical System

3.54.1 Power requirements

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The vehicle master run switch shall not carry circuit operating loads but shall control load carrying relays or magnetic switches.

3.54.2 Wiring General Requirements

Wire sizes, insulation requirements, materials, shielding methods, and identification of wire and cable used for primary, auxiliary, control, and communications applications shall be based on the current carrying capacity, voltage drop, mechanical strength, temperature, and flexibility requirements of SAE J1292.

All wire and cable insulation shall meet the flame and smoke test requirements of this specification and be substantially free of halogens. The contractor shall mark each wire every foot by wire type and each wire end with a function code using a scheme.

Braided copper wire shall be used in all ground strap applications. Flexible stranded copper wire is acceptable in other applications. All conductors of multi-conductor cables shall be terminated. The shields on multi-conductor low voltage control cables shall be terminated on one end only to minimize the effects of ground-loops.

3.54.3 Wiring and Terminals

All lamp sockets shall be of two-wire design with disconnects to eliminate corrosion or ground problems. To facilitate servicing, all lamp wires shall have leaders of at least 6 inches.

All exposed wiring between major electrical components and terminations shall have double electrical insulation, shall be waterproof, and shall conform to specification requirements of SAE Recommended practice J1127 and J1128. The requirement for double insulation shall be met by sheathing all insulated wires and harnesses with non-conductive rigid or flexible conduit.

Except as interrupted by the master battery disconnect switch, wiring shall be grouped and numbered and/or color coded and shall conform to specification requirements of SAE Standard J1127-Type SGT, SGX, or GXL and SAE recommended practice J541. Wiring shall be color coded and/or coded with a configuration code to determine circuitry. Wiring numbers shall be marked every 6 inches. Insulation shall permit ease of replacement. All wiring harnesses over 5 feet long and containing at least 5 wires shall include 10 percent excess wires for spares.

Wiring in exposed areas shall be double insulated. Double insulation shall be maintained as close to the terminals as possible. The requirement for double insulation shall be met by sheathing all wires and harnesses with non-conductive rigid or flexible conduit. Strain-relief fittings shall be provided at points where wiring enters all electrical components. Grommets of elastomeric material shall be provided at points where wiring penetrates metal structures outside of electrical enclosures and all other apertures in the vehicle. Any clamps used throughout the electrical system shall be stainless steel and of aircraft-type quality and shall be "dipped" and shall be spaced no greater than 24 inches on center. Individual tie wraps may be used to bundle wires and cable, but shall not be used to support the bundles; however, tie wrap type clamps are approved for use where it is the OEM commercial practice to utilize such

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clamps for support. Wiring supports shall be non-conductive. Precautions shall be taken to avoid damage from heat, water, solvents, or chafing. All wiring shall be kept at least 15cm (6 inches) away from heat sources, unless protected by appropriately rated reflective shields, heat resistant tubing, or high temperature loom. Wiring length shall allow for replacement of end terminals twice without pulling, stretching, or replacing the wire. Except for large wires such as battery cables, terminals shall be crimped to the wiring and may be soldered only if the wire is not stiffened above the terminal and no flux residue remains on the terminal. Terminals shall be corrosion-resistant, full-ring type or interlocking lugs with insulating ferrules.

Connectors shall be common, and self-aligning. Some suitable manufacturers include Delphi/Packard, Amphenol, Amp, Molex and Deutsch.

3.54.4 Junction Boxes and Fittings

Boxes, covers, and fittings may be constructed of aluminum, fiberglass or galvanized ferrous metals. Screws and other hardware shall be made of stainless steel.

3.54.5 Cleating

Any low voltage cables and wiring not installed in conduits shall be cleated by split block cleats of molded neoprene rubber or an approved equivalent. A nonflammable insulating material with a durometer reading of 50 to 60 shall be used for cleating. The holes in the cleat shall be sized for the individual wires and cables.

Each cleat shall spread the bolt clamping force over the entire length of the cleat. Bolts shall have lock nuts of approved design. Cable and wiring using cleating shall be supported to the bus vehicle body at least every 24 inches, except cleating supporting conduit shall be spaced as required to prevent conduit sag and movement.

3.54.6 Splicing and Taping

Splicing and taping shall not be allowed unless approved by the Government

3.54.7 Circuit Protection

All main and branch circuits shall be protected by circuit breakers or fuses sized to the requirements of the load. The circuit breakers or fuses shall be easily accessible for authorized personnel. Fuses shall be used only where circuit breakers are not practicable. Any manually re-settable circuit breakers shall provide visible indication of open circuits. Circuit breakers shall meet the requirements of SAE J553 or SAE J1625, or UL489/UL1077.

Circuit breakers or fuses shall be sized to a maximum of 15 percent larger than the total circuit load current. The current rating for the wire used for each circuit shall exceed the size of the circuit protection being used.

Handles shall indicate ON, OFF, and TRIPPED positions, or alternatively, only ON, OFF if there is no "TRIPPED" position. Fail-safe automatic reset circuit breakers shall be employed when it is necessary to quickly re-establish circuit continuity when that portion of the wiring has been subjected to an overload condition. Circuit breakers shall be molded-case type, single- or multi-pole, with frame size suitable for continuous current and interrupting duty. Each pole shall be equipped with a trip mechanism consisting of an inverse time element for overload

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protection and an instantaneous magnetic element for short circuit protection. Each pole shall be equipped with adequate means of arc extinction to prevent flashover. Multi-pole breakers shall operate contacts simultaneously.

Breaker current rating shall be clearly visible after installation.

3.54.8 Grounding

Dedicated insulated wires and/or copper braided straps shall be used for all electrical equipment, except where it can be demonstrated that dedicated grounds are not feasible or practicable. One ground plane may be the bus vehicle body and framing. Ground conductors and terminals can be clustered and shall be accessible for servicing. Grounding terminals shall be attached to ground conductors in conformance with SAE J163. No more than four (4) terminals shall be connected to a ground bolt. Grounds shall not be carried through water piping, hinges, bolted joints (except those specially designed as electrical connectors), or power plant mountings. Electrical equipment shall not be located in an environment that will reduce the performance or shorten the life of the component or electrical system. Major wiring harnesses shall not be located under the bus floor, and under floor wiring shall be eliminated to the extent practicable. Wiring and electrical equipment necessarily located under each bus shall be insulated from water, heat, corrosion, and mechanical damage, and shall be contained in sealed rigid or flexible conduit.

3.54.9 Electrical Components

Grounds, using dedicated insulated wires and/or copper braided straps, shall be connected directly to the vehicle chassis, not through hinges or panels. Ground conductors and terminals can be clustered and shall be accessible for servicing. Grounding terminals shall be attached to ground conductors in conformance with SAE J163. Grounds shall not be carried through water piping, hinges, bolted joints (except those specially designed as electrical connectors), or power plant mountings. Electrical equipment shall not be located in an environment that will reduce the performance or shorten the life of the component or electrical system. Major wiring harnesses shall not be located under the bus floor, and underfloor wiring shall be eliminated to the extent practicable. Wiring and electrical equipment necessarily located under each bus shall be insulated from water, heat, corrosion, and mechanical damage, and shall be contained in sealed rigid or flexible conduit.

3.54.10 Low Voltage Battery Disconnect Switch

A battery disconnect switch shall be furnished and located at an easily accessible location by the battery box. Buses must be equipped to allow low voltage systems to be powered off so that the battery is not drained, with infrequent high-voltage charging during the non-operational season.

3.54.11 Modular Design

Design of the electrical system shall be modular so that each major component, apparatus panel, or wiring bundle is easily separable with standard hand tools or by means of connectors. Each module, except the main body wiring harness, shall be removable and replaceable in less than 30 minutes by a 3M mechanic. Power plant wiring shall be an independent wiring

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module. Replacement of the rear compartment wiring module(s) shall not require pulling wires through any bulkhead or removing any terminals from the wires.

3.54.12 Microprocessor/PLC Based Systems

The primary purpose of the multiplexing system is control of components necessary to operate the vehicle. This is accomplished by processing information from input devices and controlling output devices through the use of an internal logic program.

Versatility and future expansion shall be provided for with expandable system architecture. The multiplex system shall be capable of accepting new inputs and outputs through the addition of new modules and/or the utilization of existing spare inputs and outputs. All like components in the multiplex system shall be modular and interchangeable with self-diagnostic capabilities. The modules shall be easily accessible for troubleshooting electrical failures and performing system maintenance. Multiplex input/output modules shall use solid-state devices to provide extended service life and individual circuit protection.

Ten percent of the total number of inputs and outputs, or at least one each at each zone location shall be designated as spares.

Multiplexing shall be centralized. A centralized system shall consist of several modules connected to form a control network. The system shall be managed by a master vehicle controller. It provides the configurability and the control required to integrate all systems on the bus. The vehicle shall be equipped with a multiplexing system.

3.55 Materials and Workmanship

Materials for the construction of the bus vehicle shall be in accordance with a generally accepted, domestic specification or cited standard (such as ASTM, AISI, SAE, other) unless the contractor obtains written approval for a substitution from the Government. All materials shall economically and safely perform and be satisfactory within their operating environment and in accordance with their intended function. Overall work quality shall conform to the best manufacturing practices in all respects. All parts and components shall be new and in no case will used, reconditioned, repaired, or otherwise obsolete parts be accepted by the Government. Any component(s) determined to be nonconforming to the requirements herein shall not be repaired or modified for use without Government approval.

Whenever a commercial material is not covered by a recognized specification or standard, the contractor shall identify the material by the commercial trademark, name, and address of the supplier. The contractor shall submit a description of the material composition for approval by the Government.

Single-source materials shall not be permitted unless approved. Approval shall be determined on a case-by-case basis. Specification equivalency/benefit data for any substitution to a cited standard shall be submitted to the Government for review and approval.

The following materials shall not be used in the construction of the bus:

- PVC;

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- Asbestos;
- Lead (in brake shoes);
- Urethane foam;
- Chlorinated fluorocarbons that may cause environmental problems or handling hazards;
- Materials that, in their normal installed state, emit products that are known to be toxic or irritative.

Bus vehicle body structural parts that are permanently covered and concealed after assembly shall not be made of copper, copper bearing aluminum alloys, brass, bronze, silver, or nickel. Foreign matter, such as shavings, chips, etc., shall be completely removed from all parts of the vehicle, its components, assemblies and subassemblies, whether hidden or exposed. Surfaces exposed to passengers, crew, or maintainers shall be smooth and free of burrs, fins, sharp edges, and dangerous protrusions.

The contractor shall keep on file material safety data sheets (MSDS) for all chemical materials (paints, solvents, adhesives, etc) used in the manufacture of the vehicles, and provide MSDS information as requested by the Government for any questionable material.

All non-metallic materials utilized in the construction of the vehicles shall be subject to the approval of the Government. The use of caulking and sealing compounds shall be minimized. Caulking and sealing compounds shall be applied in accordance with the manufacturer's instructions and recommendations, shall be non-staining, and shall be supplied in colors closely matching those of adjacent materials and surfaces. Caulking compound application shall be fully compatible with subsequent paint application (as applicable).

Threaded aluminum fasteners shall not be used, except in tapped holes in solid aluminum structures, subject to approval and review by the Government.

All name and rating plates shall be permanently attached using mechanical fasteners. Exceptions may be made for small components and circuit boards. All materials shall be new and of recent manufacture. Material which is found defective and repaired cannot be used unless specific approval is granted by the Government.

The contractor shall maintain records that trace all materials to their manufacturers, and shall help verify compliance with quality standards specified or cited in this specification. This information shall be made available to the Government upon request.

3.55.1 Units of Measure

Each vehicle system shall be designed and manufactured to a single standard. U.S. inch and ISO metric parts shall not be mixed within components.

3.55.2 Storage of Materials

All stored material subject to corrosion shall be protected by waterproof covers or coatings. Equipment covers, cable entrances, and openings shall be closed to prevent ingress of water or

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dirt. All dated material shall have the expiration date clearly marked. Expired material shall not be used. Rejected or damaged material shall be clearly marked and separately stored.

3.55.3 Fasteners

The contractor shall ensure the proper application of all non-threaded and threaded fasteners. Torque marks or stripes extending from the secured hardware shall be applied to all safety related hardware, including door, and brake equipment bolts. In cases of suspect workmanship, the Government may require non-safety related hardware to have a tightening indication present which is not required to extend onto surrounding surfaces.

3.55.4 Threaded Fasteners

The number of different sizes and styles of fasteners used shall be minimized and shall meet the following requirements:

- All fasteners shall be stainless steel, dichromate, or zinc plated steel, depending on the specific application;
-
- No protruding screws, mounting bolts, or similar items shall be permitted on the exterior of the vehicle;
- On the vehicle interior, all exposed screws shall be stainless steel with flat or oval heads;
-
- Fasteners not exposed to passengers on the vehicle interior shall be of stainless steel or zinc-plated steel;
-
- Use of threaded inserts or special fasteners shall require approval;
-

3.55.5 Locking Requirements

Locking requirements for threaded fasteners shall be in accordance with the following:

- All threaded fasteners shall be self-locking or provided with locking devices;
- Locking devices shall be either lockwire, lock-washers or locknuts as appropriate for the application or service;
- Lockwire, if used, shall be stainless steel;
- Locknuts shall be of the deformed thread prevailing torque type;
- Bolts for use with locknuts shall not be heated or drilled for cotter pins;
- All locknuts shall comply with the Industrial Fasteners Institute requirements regarding to locking ability.

3.55.6 Torque Marking/Indexing

The contractor shall ensure the proper application of all threaded fasteners. Torque marks or stripes extending from the secured hardware shall be applied to all safety-related hardware, including bus vehicle body, door, and brake equipment bolts.

3.55.7 Riveting

Rivet holes shall be accurately located and aligned. Rivets shall completely fill the holes. Hand-driven steel rivets shall be driven hot and shall completely fill the holes. Mechanically driven rivets may be driven cold. Use of blind rivets shall require Government approval. All rivets shall be of a solid center type. External rivet heads shall be concentric with the body of the rivets and free from rings, pits, burrs, and fins. Surfaces exposed to passengers, driver(s), or maintenance personnel shall be smooth and free of burrs, fins, sharp edges, and dangerous protrusions.

3.55.8 Dissimilar Metal Treatment

Isolating and moisture-proofing materials, appropriate to the materials being joined, shall be used.

Areas exposed to corrosive fluids or cleaning solutions shall be protected with coatings resistant to those fluids. The contractor shall be responsible for verifying that all such areas are protected through communications with the Government.

Except as otherwise indicated, all aluminum exposed to view in finished work in the interior of the vehicle shall have a protective anodized coating. Surfaces of aluminum alloy parts secured to ferrous parts shall be protected with one-part polysulfide or silicone sealant used as joint compound, or with joint material that is non-hygroscopic and is free from chlorides and heavy metal ions.

3.55.9 Welding

All welding practices shall be in accordance with the applicable requirements and recommendations of the American Welding Society (AWS), as contained in the latest revisions of the "Structural Welding Code" (AWS D1.1), "Aluminum Welding Code" (AWS D1.2), "Structural Welding Code - Sheet Steel" (AWS D1.3), "Recommended Practices for Resistance Welding" (AWS C1.1), and the AWS "Welding Handbook" (AWS WHB). Where non-AWS welding practices or processes will be used, such as Canadian Standards Association (CSA), the contractor shall demonstrate equivalence for Government review and approval.

The contractor shall be responsible for the quality of all welding and brazing, including the welding and brazing of its suppliers and sub-contractors. Prior to welding, all surfaces shall be thoroughly cleaned to remove corrosion, rust, scale, slag, grease, oil, water, paint, and other foreign materials. Parts to be joined by welding shall be supported and held in position by tables, jigs, or fixtures to prevent warping.

All weld quality shall be in accord with acceptable weld criteria as defined in AWS or CSA welding codes.

The contractor shall have an on-going quality control program of inspection, non-destructive or destructive testing to insure quality welds. The contractor shall provide for Government review and approval complete documentation as to how its welders are certified and monitored. Complete documentation shall further be provided describing, in detail, the type of weld testing performed, frequency of the tests, and actions taken if defective workmanship is

discovered. Documentation shall also be provided on the testing and monitoring of the welding devices utilized.

3.56 Air and Hydraulic Hoses

Air or hydraulic hose applications shall not be permitted in locations where adequate visual inspections cannot be made. Air hose installations shall be located/arranged in such a manner as to prevent accidental cross connections to other hoses located in the same general area. Hose installations shall be such that kinking, rubbing, straining, and unnecessary swinging are precluded. Routing that requires other piping or cables as the sole means of support shall not be accepted.

3.57 Air Pressure Vessels

Brake air reservoirs shall comply with SAE J10 Automotive and Off-Highway Air Brake Reservoir Performance and Identification Requirements.

3.58 Bearings and Lubrication

All bearings and lubricants shall be readily available in the United States. Standard grease fittings or plugs shall be provided for all bearings not internally splash- or bath-lubricated. All rotary shafts shall be supported by cylindrical or tapered roller bearings where practicable. Ball bearings may be used, subject to approval.

Bearings subject to atmospheric or liquid contamination shall be sealed by labyrinth, lip, or face seals. Bearings that are not splash or bath lubricated shall be provided with standard grease fittings and drain plugs or pressure-release devices for relubrication. Ball-bearings of 25 mm [1-in] shaft size and smaller may be factory lubricated-for-life, subject to approval. Bearings shall be installed and removed without major disassembly of related components, and in no case shall thrust be carried across the rolling elements in pressing a bearing into its seat.

Sleeve bearings shall be used for shafts with rotary motion of less than one full revolution. Sleeve bearings shall be adequately lubricated. Sleeve bearings supporting ferrous shafts shall be composed of bronze, brass, or aluminum alloys as approved. Sleeve bearings may be used to support rotary shafts if space limitations preclude the use of anti-friction bearings.

Self-lubricated bushings (sintered metal) shall be used in accordance with the manufacturer's recommendations, but shall not be used for shafts with speeds greater than 500 rpm.

All lubricants shall be products approved by the supplier of the parts on which the lubricant is to be used. All lubricants shall, as a minimum, conform to applicable ANSI and ASTM specifications. Multi-purpose lubricants shall be used where possible.

3.59 Flammability and Smoke Emissions

Materials used in the vehicle shall comply with the flammability, smoke emission, toxic gas, and fire retardation requirements specified herein. Those materials and products generally recognized to have high toxic products of combustion shall not be used. Materials used in the vehicles shall be zero halogen in addition to meeting the low-smoke requirements specified below. Unless specified below, all materials and construction shall meet applicable NFPA

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requirements. All materials used in the construction of the passenger compartment of the bus vehicle shall meet the requirements of FMVSS 302. In addition, with the exception (only) of the following list of materials, all materials used in the construction of the passenger compartment shall be tested in accordance with and meet the performance criteria in the recommended Fire Safety Practices as defined in FTA Docket 90-A, dated October 20, 1993.

Only the materials listed below are not required to meet FTA Docket 90-A:

- Adhesives, sealants and undercoating, including RTV and PPG undercoating
- Front and rear fiberglass caps
- HVAC air return grill-ABS material
- Polycarbonate sheets for driver' barrier and door modesty panels
- Driver's dash box and side console-ABS material.
- Window and door rubber seals
- Flooring material adhesive to floor
- Rear compartment and under floor noise insulation
- Passenger compartment light covers
- Driver seat components (depending on specifications)
- Passenger seat handles if not stainless steel

Materials proposed for the construction of the bus which are listed in the aforementioned FTA Docket 90 shall be test certified according to the defined procedures and performance criteria in FTA Docket 90-A.

The Government will review and approve contractor submitted material certification(s) proving that the materials to be utilized in the construction of the proposed bus meets the aforementioned FMVSS and/or FTA Docket 90 requirements. All testing shall be conducted by a recognized, independent laboratory testifying to material conformance. The contractor shall provide to the Government complete test report results prior to shipment of the first production bus vehicle.

3.60 General Workmanship

A. Buses shall be free from defects that may impair their serviceability or detract from appearance.

B. All components will be new. Defective components shall not be furnished.

Parts, equipment, and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished without the approval of the Government. Component parts and units shall be manufactured to definite standard dimensions with proper fits, clearances and uniformity. General appearance of the vehicle shall not show any evidence of poor workmanship.

C. The following shall be reason for rejection, as examples:

1. Rough, sharp or unfinished edges, burrs, seams, corners, and joints.
2. Non-uniform panels and edges that are not radiused, beveled, etc.
3. Paint runs, sags, orange peel, "fish eyes," etc., and any other imperfection or lack of complete coverage of paints or coatings or wrap bubbles, misalignment, unadhered edges, etc..

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4. Body panels or components that are uneven, unsealed, or contain cracks, dents or have voids.
5. Misalignment of body fasteners, glass, viewing panels, light housings, other items with large or uneven gaps, spacing etc. such as door, body panels and hinged panels.
6. Improperly fabricated and routed wiring or harnesses, and electrical connections.
7. Improperly supported or secured hoses, wiring harnesses, mechanical controls etc., including interference with other components.
8. Interference of chassis components, body parts, doors, etc.
9. Leaks of any gas, or fluid lines (air conditioning, coolant, oil, etc.).
10. Noise, panel vibrations etc.
11. Inappropriate or incorrect use of hardware, fasteners, components, or methods of construction.
12. Incomplete or improper welding, riveting or bolting.
13. Lack of uniformity and symmetry where applicable.
14. Loose, vibrating, abrading body parts, components, subassemblies, hoses, wiring harnesses or trim.
15. Improper body design or interface with the chassis substructure that could cause injury during normal use or maintenance, and which fail to provide access to perform routine or mandatory repairs or maintenance on vehicle electrical and mechanical systems. In addition, the improper combination of options which by their combination and installation are inherently incompatible with regard to function or safety.
16. Sagging non-form fitting upholstery or padding, holes, tears, discoloration, etc.
17. Visual deformities and equipment malfunctions.
18. Unsealed appurtenances or other body components, gaskets, etc.
19. In addition, any deviation from specification requirements or any other item, whether or not stipulated herein, that affects form, fit, function, finish, durability, reliability, safety, performance or appearance shall be cause for rejection.

3.61 Manuals and Catalogs

Driver's instruction manuals shall be provided to the Government in the quantities as follows:

- Driver's Instructional Manual – 2 copies per bus

The Driver's Instruction Manual shall contain all information needed for the optimum operation of the bus vehicles; including general vehicle familiarization material, location, function and operation of all controls, indicators, switches, and emergency procedures and trouble diagnoses.

3.62 Warranty Requirements

The bus shall be reliable in operation and shall operate with a mean distance between failures (MDBF) of 10,000 miles or greater. MDBF is defined as the miles traveled between a failure that requires a road call (the vehicle cannot complete its normal service). The bus shall operate with no incidents requiring outside repair or vehicle down time within the first two calendar years of operation. Any repairs performed under warranty shall be resolved within 5-business days with an initial response to a warranty issue received within 2-business days. All travel, parts, and labor costs for warranty related repairs are to be borne by the Offeror.

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Warranties in this document are in addition to any statutory remedies or warranties imposed on the Offeror. Consistent with this requirement, the Offeror shall warrant and guarantee to the Government each complete bus vehicle and specific subsystems and components according to the following provisions. All component warranties shall be manufacturer's warranties furnished by the manufacturer of the component.

The Offeror shall, at its own expense, have one or more competent representative(s) available on request to assist the government staff in the solution of engineering or design problems that are within the scope of the specifications and that may arise during the warranty period.

Complete Bus:

The complete bus shall be warranted and guaranteed to be free from defects or related defects for 3 years or 150,000 miles, whichever comes first, beginning on the date the NPS places the bus into service.. During this warranty period, the bus vehicles shall maintain their structural and functional integrity. The warranty shall be based on regular operation of the bus vehicles under the operating conditions prevailing in Government service.

Chassis Structure:

The chassis structure is composed of all components that are welded together to form the main frame and body structure. Bolted-on components, unsprung suspension components, and operating hardware are considered add-ons and therefore are not a part of the basic body structure.

The chassis structure, including structural elements of the suspension attached to the body structure but excluding suspension walking beams, shall be warranted against corrosion failure and/or fatigue failure for a period of 12 years or 500,000 miles, whichever comes first.

Subsystem and Components:

Specific subsystems and components shall be warranted and guaranteed to be free from defects and related defects for the times or mileages given in Table 0-1 below, beginning on the date the NPS places the bus into service. . Extended warranties shall be purchased by the contractor if required to meet the required warranty years/mileage in Table 0-1. The contractor shall furnish the manufacturer's base or standard warranty certificates for all warrantable items, and proof of purchase from the manufacturer of extended warranties, all to be included with each bus vehicle. Copies of base or standard warranty certificates and proof of purchase of extended warranties shall be furnished to the government inspector prior to acceptance of the bus by the inspector and sent to the contracting officer within 10 days after acceptance of the bus. The proof of purchase of extended warranties shall consist of at least one of the following:

- Extended warranty certificate with specific item serial number
- Invoice for extended warranty referencing specific serial number and proof of payment

Table 0-1: Subsystem and Component Warranty

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Item	Description	GSA Baseline		Offeror's warranty	
		Years	Mileage	Years	Mileage
1	Bumper to bumper	3	150,000		
2	Monocoque Structure	12	500,000		
3a	Battery ESS- base materials/w workmanship	12	unlimited		
3b	Battery ESS-state of health	6	unlimited		
3c	Battery-ESS	12	No more than 70% degradation		
4	Traction Motor and Electrical Systems	5	250,000		
5	Corrosion protection	7	350,000		
6	HVAC	3	unlimited		
7	Exterior Vinyl Graphics	4.5	unlimited		

Voiding of Warranty:

The warranty shall not apply to any part or component of the bus vehicles that has failed as a direct result of misuse, negligence, accident, or that has been repaired or altered in any way so as to affect adversely its performance or reliability, except insofar as such repairs were in accordance with the Offeror's maintenance manuals and the workmanship was in accordance with recognized standards of the industry.

The warranty on any part or component of the bus vehicles shall also be void if the Government fails to conduct normal inspections and scheduled preventive maintenance procedures on the same part or component substantially as recommended in the Offeror's maintenance manuals and such failure by the Government is the sole cause of the part or component failure.

Exceptions to Warranty:

The warranty shall not apply to scheduled maintenance items and items furnished by the Government, except insofar as such equipment may be damaged by the failure of a part or component for which the Offeror is responsible.

Detection of Defects:

If the Government detects a defect within the warranty periods, it shall promptly notify the Offeror's representative. Within 5 working days after receipt of notification, the Offeror's representative shall either agree that the defect is in fact covered by warranty, or the offeror shall examine at Government property.. At that time the status of warranty coverage on the subsystem or component shall be mutually resolved between the Government and the Offeror. Work shall commence within 10 working days after receipt of notification to the Offeror.

Scope of Warranty Repairs:

When warranty repairs are required, the Government and the Offeror's representative shall agree within 5 working days after notification of a detected defect on the most appropriate course for the repairs and the exact scope of the repairs to be performed under the warranty. If no agreement is obtained within the 5-day period, the Government reserves the right to commence the repairs and seek reimbursement from the offeror.

Repair Parts and Service

Continuous operation of the buses is of utmost importance. Parts such as body panels, window assemblies excluding glass panels and windshields, doors, seats, seat covers, motors, and transmissions shall be shipped within 30 days from order. Parts shall be available for a period not less than 12 years from the date of delivery.

3.63 Testing and Quality Assurance

3.63.1 Water Spray Test

The OEM's standard water spray test shall be used, subject to Government approval.

Testing shall be conducted in accordance with the bus manufacturer's standard testing procedures on all buses ordered. Offeror shall supply all Testing, Quality Assurance, and documentation procedures used at the factory.

Evidence of water leakage into passenger compartment shall be cause for rejection until leaks are corrected.

3.64 Verification Requirements

The contractor shall verify compliance to the requirements herein prior to first bus production release. Verification may be in the form of contractor production line bill of material,

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drawings, instructions, test results, vendor documentation, certifications, or design analytical data. The contractor shall not proceed with production before receiving agreement from the Government that verification of compliance is satisfactory. Acceptance of designs by the Government does not relieve the contractor from the contractor's responsibility to meet all the requirements and specifications herein, whether reviewed by the Government or not. It shall be the contractor's responsibility to assure all production documentation results in the production of vehicles that are in compliance with these requirements and specifications.

PWO Provide With Offer

PAC Post Award Conference

Contract Award + 5 Days

VDR Verification Design Reviews

Receipt of Order + 30, + 60 Days

FPVI First Production Vehicle Inspection

BOM Bill of Material

The type of verification to be submitted for acceptance is listed below:

ET Engineering Test- to be performed by the contractor and results accepted by the Government. Engineering tests are tests resulting in data output from engineering instruments used to take measurements of parameters such as temperatures, sound levels, etc.

QAPVT Quality Assurance Performance and Verification Testing- to be performed by the contractor and verified by the contractor to the Government. QAPVT tests are functional tests that do not involve instrumentation. The results are observable and recorded as pass/fail, i.e. wheelchair lift platform functions as intended; lights function as intended, etc. Systems or components that fail shall be annotated with the corrective action taken and retested until passed.

The contractor shall, within 45 days after receipt of order, submit detailed plans for the performance of all ET tests. These plans, at a minimum, shall include all instruments to be used, their calibration certification, and the procedures to be followed.

The Government may accept designs at the PWO, VDR, or FPVI. These acceptances shall constitute design freezes and enable the contractor to proceed with production activities relative to the accepted designs. The design freeze shall be effective for the duration of the production of the entire quantity of buses being procured.

3.65 GSA Form 1398

Contractors shall affix one copy of GSA Form 1398, GSA Purchased Vehicle, fully completed, to the right or left front door lock face or door post after the pre-delivery inspection is made at the manufacturer's plant. All marks on windows and other labels (except the Monroney label and labels cautioning against drained transmission, crankcase, and rear axle) shall be removed. Copies of GSA Forms 1398 are available from the Contracting Officer.

NOTE TO OFFEROR: Data shown in the solicitation schedule of items may be utilized to satisfy the requirements for the following to be shown on GSA Form 1398:

(1) Receiving Agency— National Park Service.

(2) Purchase Order No.—Use the five-digit "case" or "file" number (often referred to as the RPN number)

4.0 Reference Information

4.1 Abbreviations and Terminology

The following is a list of abbreviations used in this Specification:

- ADA Americans with Disabilities Act
- ANSI American National Standards Institute
- ASHRAE American Society of Heating, Refrigerating and Air Conditioning Engineers
- ASE Automotive Service Excellence
- ASTM American Society for Testing and Materials
- FTA Federal Transit Administration
- FMCSR Federal Motor Carrier Safety Regulations
- FMVSS Federal Motor Vehicle Safety Standards
- JIC Joint Industrial Council
- Government General Service Administration
- OSHA Occupational Safety and Hazard Administration
- SAE Society of Automotive Engineers
- UL Underwriter's Laboratory
- USEPA United States Environmental Protection Agency

4.2 Definitions

The following are definitions of special terms used in this Specification:

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- Approach Angle - The approach angle is the angle measured between a line tangent to the front tire static loaded radius arc and the initial point of structural interference forward of the front tire to the ground.
- Breakover Angle - The breakover angle is the angle measured between two lines tangent to the front and rear tire static loaded radius and intersecting at a point on the underside of the vehicle that defines the largest ramp over which the vehicle can roll.
- Bus Body and Chassis OEM (Bus body and chassis Original Equipment Manufacturer)- The manufacturer responsible for all of the following:
 1. The design and manufacture of the bus body and chassis, including, but not limited to, the bus undercarriage and bus body;
 2. The final assembly of undercarriage, body, drive, and components into a complete and functional bus;
 3. The warranty of all components, systems, and subsystems (excluding tires);
 4. The assigning of the DoT VIN (Vehicle Identification Number).
- Class of Failures - Classes of failures are described below.
- Class 1: Physical Safety - A failure that could lead directly to passenger or driver injury or represents a severe crash situation.
- Class 2: Road Call - A failure resulting in an en route interruption of service. Service is discontinued until the bus is replaced or repaired at the point of failure.
- Class 3: Bus Change - A failure that requires removal of the bus from service during its assignments. The bus is operable to a rendezvous point with a replacement bus.
- Class 4: Bad Order - A failure that does not require removal of the bus from service during its assignments but does degrade bus operation. The failure shall be reported by operating personnel.
- Contractor - Person or persons, firm, partnership, corporation, or combination thereof that has entered into the procurement contract with the Government.
- Curb Weight - Weight of vehicle, including maximum fuel, oil and coolant; and all equipment required for operation and required by this specification, but without passengers or driver.
- dBA - Decibels with reference to 0.0002 microbar as measured on the "A" scale.

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- Departure Angle - The departure angle is the angle measured between a line tangent to the rear tire static loaded radius arc and the initial point of structural interference rearward of the rear tire to the ground.
- Fireproof - Materials that will not burn or melt at temperatures less than 2,000 degrees Fahrenheit.
- Fire-Resistant - Materials that have a flame spread index less than 150 as measured in a radiant panel flame test per ASTM-E 162-90.
- Free Floor Space - Floor area available to standees, excluding the area under seats, area occupied by feet of seated passengers, areas forward or to the side of standee lines, and any floor space indicated by manufacturer as non-standee areas such as, the floor space “swept” by passenger doors during operation. Floor area of 1.5 square feet shall be allocated for the feet of each seated passenger that protrudes into the standee area.
- Gross Load – Load based on the number of seated and standee passengers, plus driver, at 150 pounds/passenger, the bus can carry without exceeding the front or rear GAWR and GVWR.
- GVW (Gross Vehicle Weight) - Curb weight plus gross load.
- GVWR (Gross Vehicle Weight Rating) - The maximum total weight as determined by the vehicle manufacturer, at which the vehicle can be safely and reliably operated for its intended purpose.
- GAWR (Gross Axle Weight Rating) - The maximum weight, as determined by the vehicle manufacturer, that can be carried on a particular axle, or set of axles, with a given rim and tire.
- Human Dimensions - The human dimensions used in this Specification are defined in SAE Recommended Practice J833.
- Industry/Service Proven – Component/System having two years of proven revenue service.
- Low Floor Bus - A bus which, between at least the front (entrance) and rear (exit) doors, has a floor sufficiently low and level so as to remove the need for steps in the aisle between the doors and in the vicinity of these doors.
- Maintenance Personnel Skill Levels - Defined below are ASE maintenance personnel skill levels used in this Specification.
 - 5M: Specialist Mechanic or Class A Mechanic Leader
 - 4M: Journeyman or Class A Mechanic
 - 3M: Service Mechanic or Class B Servicer

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- 2M: Mechanic Helper or Bus Servicer
- 1M: Cleaner, Fueler, Oiler, Hostler, or Shifter

Note: Whenever a specific time is indicated to access components or complete a task, it is assumed the vehicle is in the location where the work is to be performed. All necessary equipment is in its correct position (tools, jacks, vehicle lifts, lighting, fluid recovery systems, etc.) and ready for use.

- Driver - Individual on board who is in charge of bus operation.
- Driver's Eye Range - The 95th-percentile ellipse defined in SAE Recommended Practice J941, except that the height of the ellipse shall be determined from the seat at its reference height.
- Seated/Standee Load – Two hundred fifteen pounds (150 lbs) for every designed passenger seating position and for the driver.
- Standee Line - A line marked across the bus aisle to designate the forward area that passengers may not occupy when the bus is moving.
- SLW (Seated Load Weight) - Curb weight plus seated load.
- Standards - Standards referenced in this Specification are the latest revisions unless otherwise stated.
- Structure - The structure shall be defined as consisting of all the components that are welded together to form the main frame (skeleton) and body construction including axle and suspension attachment points.
- Wheelchair - A mobility aid belonging to any class of three or four-wheeled devices, usable indoors, designed for and used by individuals with mobility impairments, whether operated manually or powered. A “common wheelchair” is such a device that does not exceed 30 inches in width and 48 inches in length measured two inches above the ground, and does not weigh more than 600 pounds when occupied.

4.3 First Vehicle Inspection (FVI) and Pre-Inspection Test Drive

The contractor shall furnish a first production vehicle for inspection by Government representatives within the time frame specified by the Contract. Production methods shall assure that subsequent vehicles are identical to the approved first production vehicle. The first production vehicle shall be complete in every respect, i.e., all components, equipment and accessories assembled and/or installed and fully functional and clean.

The contractor shall inspect the bus for compliance to all the requirements herein, including all workmanship requirements, prior to Government inspection. The purpose of the Government inspection is primarily to verify the contractor's inspection and corrective actions. Prior to the Government inspection, the contractor shall furnish the Government inspector verification that the inspection has been made. This verification shall, at a minimum, consist of the following:

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- In –process and final inspection documentation
- Non-compliances found during in-process and final inspection
- Disposition of all non-compliances and rejections (corrective actions)
- Test results as required herein
- Performance Assurance and Verification results as required herein
- Test drive results, inspection, and corrective actions
- Deficiencies discovered by the Government inspector on the vehicle presented for first vehicle inspection may subject the contractor to reinspection charges.

It shall be the responsibility of the contractor to verify to the Government Inspector that all tests and inspections have been made and that the vehicle presented meets all the design, performance, material, and workmanship requirements herein.

After approval of the first production vehicle, no substitution of materials, components, or assemblies will be permitted except by modification to the contract. Prior to the first vehicle inspection, the bus shall be driven on a road test by the contractor for a minimum of 20 miles. A portion of this test may be simulated on a chassis dynamometer. The road test shall be comprised of the following:

- Prior to the road test, diagnostics shall be obtained and printed from the ESS controller. Any existing fault codes shall be noted, diagnosed, corrected, and cleared.
- 20 miles of continuous state or interstate highway driving at speeds of 50-65 MPH. The bus shall be stopped a minimum of 5 times at highway turnarounds and/or other appropriate and safe places.
 - Each stop shall include one door opening/closing cycle of both doors.
 - The wheelchair ramp shall be activated at each stop.
- At the end of the test drive, another diagnostic reading from the ESS controller shall be printed. Any fault codes generated from the test drive shall be noted, diagnosed, the cause(s) corrected, and cleared.
- The bus shall be inspected after the test drive for the following:
 - Fluid leaks- (defined as fluid seepage resulting in a drop being formed) shall be recorded and corrected.
 - Pneumatic, hydraulic, HVAC, or coolant hose abrasion against bus structure or components. Hose routing deficiencies from this inspection shall be recorded and corrected.
 - Components that loosened, rattled, or otherwise caused reason for notation during the test drive. Corrective action shall be taken and recorded.

The contractor shall document all incidents, malfunctions, and any unusual bus performance noted during this test drive, plus the quantity of any fluids added during the drive. All other corrective actions resulting from the 20-mile test drive shall be noted. Prior to acceptance by the Government, all fault codes shall be corrected and cleared from the ECM.

4.4 Production Vehicle Final Inspection and Test Drive and Water Spray Test
Each production bus shall undergo a final inspection by the GSA Industry Operations Analyst (IOA).

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Prior to inspection and acceptance by the Government, each production vehicle shall be road test driven for a minimum of 20 miles as required in section 4.3 above. All contractor inspection and verification of contractor inspection requirements in section 4.3 above and water spray tested as required in section 3.63 shall apply to each production bus. Deficiencies discovered by the Government inspector on any vehicle presented for final inspection may subject the contractor to reinspection charges.

4.5 Optional Requirements

The following are descriptions of options, the order quantities are independent of the quantity of buses.

4.5.1 Option PSM Parts and Service Manuals

If option PSM is selected, the following shall be furnished as one set for the entire order (not per bus):

- Preventive Maintenance Manual – two copies, one with laminated pages.
- Parts Catalog – two copies, one with laminated pages.
- Service Manual-two copies, one with laminated pages, consisting of the following:
 - Electrical Schematics Manual
 - System Interface Diagrams-Electrical
 - Integrated Wiring Diagrams
- Major sub-contractor manuals (motors, transmission, wheelchair lift, HVAC)
- All publications submitted as hard copies shall, wherever possible, also be submitted in electronic format (PDF) in digital media.

Preventive Maintenance Manual

The Preventive Maintenance Manual shall enable the maintainer to perform the periodic inspection and preventive maintenance tasks, including schedules/intervals. These shall include all lubrication and inspection requirements for all apparatus requiring such work on a periodic basis to maintain the bus vehicles in satisfactory working order.

The Diagnostic and Troubleshooting section of the manual shall enable the maintainer to troubleshoot, adjust and complete running repair for all systems on the bus vehicles. Running repair is the diagnosis and correction of any subsystem or component malfunction by adjustment, repair or replacement in order to return the vehicle to service in a reasonably short period of time. The section shall include a general description and operation of each system, block diagrams, signal flow diagrams, detailed schematics with narrative and functional wiring and piping diagrams.

The manual shall contain a detailed description of components of the bus to enable a maintainer to service, maintain, and repair the vehicle. The manual shall include a complete systematic procedure for troubleshooting and long term periodic maintenance requirements for all components.

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Parts Manual

The Parts Manual shall enumerate and describe components with its related parts and contractor part numbers. Cut-away and exploded drawings shall be used to permit identification of all parts not readily identified by description. Each part or component shall be identified as being part of a functional group.

An important aspect of the parts catalog shall be the complete itemization of all servicing materials (oils, paints, special compounds, greases, other) required on the bus vehicles and the component requiring its use.

Sub-contractor/supplier manuals shall be purchased and may be furnished “as is”. Within the manuals, the bus vehicles shall be treated as a whole and not as a grouping of disassociated parts from various suppliers. It shall be the responsibility of the bus OEM to insure that all of the suppliers' subsystems are presented in sufficient detail to present a complete and clear picture of the whole vehicle and that terms and functional designations of wires and components are consistent throughout. The material in all manuals shall be identically organized and indexed with compatible numbering.

The bus manufacturer shall provide support access via telephone, 7 days x 24 hours

4.5.2 Option DIAG Diagnostic Software and Hardware

If option DIAG is selected, the following shall be furnished for the order (not per bus):

- One set of multiplex electrical system service and diagnostic software
- One hand held multiplex diagnostic tool, if available from the multiplex system manufacturer.
- Maintenance/diagnostic ports shall be provided on each bus

4.5.3 Option OTRN Driver Training

If option OTRN is selected, the contractor shall provide a minimum of 16 hours of driver training necessary for the ordering agency to safely and reliably operate the bus. Training sessions shall include training on all bus systems, subsystems, and included optional systems.

4.5.4 Option MTRN Maintenance Training

If option MTRN is selected, the contractor shall provide a minimum of 40 hours of maintenance training to the ordering agency's maintenance personnel covering, at a minimum, motors, batteries, electrical, HVAC, fuel system, and contractor proprietary system maintenance training. The bus manufacturer shall furnish documentation for first responders and park personnel on how to disable the high voltage battery system and any other safety protocols or information

4.5.5 Option ESSE Energy Storage System Extended Range

If option ESSE is selected, the contractor shall provide an energy storage system with extended range. The offer shall provide details and ratings of their proposed solution.

4.5.6 Option EVSE Electric Vehicle Supply Equipment

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Offerors may propose multiple solutions for this option, various makes/models/capacities/dispenser type/features that are compatible with the offered bus.

The offeror shall identify the respective makes and models of the EVSE configurations offered. The offeror shall provide the EVSE manufacturers' installation requirements and instructions. Installation of the EV charging stations by the Government shall not impede/limit/terminate any commercially offered warranties.

If option EVSE is selected, the contractor shall provide a stationary charging station conforming to applicable safety and industry standards.

The EVSE shall include a connector compatible with the offered bus; shall provide a minimum output of 60 kW; shall be capable of operating through cellular/wi-fi/fiber optic/internet capabilities, shall operate in an environment in temperatures between -30 C to +50C ambient; and shall include one year of data capture and storage capabilities.

The EVSE components, assemblies, and accessories shall be delivered FOB destination. All items shall be as represented in the manufacturer's technical data book. Special or mounted equipment shall be as represented in the equipment manufacturer's technical data.

The EVSE shall be listed, certified, or classified, to the applicable UL Standards, by a Nationally Recognized Testing Laboratory (NRTL) in accordance with 29 CFR 1901.7.

The EVSE is intended for use in accordance with the National Electrical Code (NEC), NFPA Article 70. The EVSE shall accept 480-VAC input. Refer to the NPS Zion EVSE Site Plan (Appendix A), provided for reference only:

All EVSE shall be suitably rated for outdoor use, either NEMA Type 3R Indoor/Outdoor Rated, NEMA 4X or equivalent with an Operating Temperature ranging from -30 C to +50C ambient.

No exterior lights shall be visible on the EVSE while the bus is charging between dusk and dawn. If required, the color temperature of the lights shall be 2300K, producing an amber glow. Lights fixtures shall be shielded to minimize their effects.

Advanced metering shall be provided. The EVSE shall be Smart Grid capable enabling demand side management. Smart Grid allows utilities to integrate EVSE meters with the utilities' Advanced Metering Infrastructure (AMI). For more information refer to <http://www.smartgrid.gov>

The EVSE shall allow NPS Zion to implement charging control strategy in order to reduce peak demand. Based upon DOE-NREL studies conducted at Zion, the daily energy use and daily charging time variations are estimated at 35 kWh per bus with daily time needed to charge buses (tied to average daily energy usage) varies from 0.5 hours to 6.5 hours. The estimated average time required to charge a bus is 4 hours.

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All networked EVSE shall be able to transmit data automatically without operator involvement to a web-based portal or network. The communication network must be a current standard sustainable network platform. In the occurrence that the communication network becomes obsolete the vendor shall ensure that the EVSE can be upgraded to accommodate the new network at no cost to the Government. There can be no loss of data. The EVSE must operate an open architecture platform that allows charging data to be communicated to EVSE owners by multiple network providers and payment mechanisms. The purpose for an open architecture platform is to prevent the charging station from becoming obsolete should the cellular network and/or software provider change at some point in the future.

If the communication network is unavailable to transmit the data, the EVSE shall store the data until network connectivity is restored and be able to transmit stored data.

The offerors shall identify any proprietary telematics, data or communications solutions in their proposals.

For each charging event the networked EVSE for each relevant Category shall collect vehicle level transactional data and transmit it to a controlled access web-based portal. This web portal will be accessed by NPS Zion. The EVSE shall collect at a minimum the following information:

1. Bus/Driver Identifier
2. Station Identifier
3. Port number (if more than one port or EVSE at a location)
4. Connect Start Date (MM/DD/YYYY)
5. Connect Start Time (HH:MM:SS)
6. Connect End Date (MM/DD/YYYY)
7. Connect End Time (HH:MM:SS)
8. Charging Start Date (MM/DD/YYYY)
9. Charging Start Time (HH:MM:SS)
10. Charging End Date (MM/DD/YYYY)
11. Charging End Time (HH:MM:SS)
12. Total Charging Time (HH:MM:SS)
13. Total actual kWh consumed

The web-based portal shall provide a means to control driver access to an EVSE. NPS Zion administrator shall be able to login to the web based portal and select specific drivers that may access specific the specific EVSE or group of EVSEs.

One (1) OEM operator manual and installation manual shall be furnished with each EVSE. A training DVD or an internet link to a training video shall be included with the offer/quote for evaluation purposes.

The training DVD and/or the training video link shall be provided with each EVSE. These training materials shall include procedures on operation of the EV charging station as well as activation and

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installation guides and FAQs/Best practices. These training materials shall also include a recommended preventive maintenance schedule and procedures for the EV charging station.

The offeror shall provide the EVSE manufacturer's commercial warranty and the commercial furnished equipment warranties, all other parts and components required herein, against parts failure or malfunction due to design, defective workmanship, and missing or incorrect parts, for a minimum period of 6 years from date of installation. If offeror receives, from any supplier or subcontractor, additional warranty on the whole or any component of the EVSE, including any prorated arrangements, or the contractor generally extends to its commercial customers greater or extended warranty coverage, the Government shall receive corresponding warranty benefits. The warranty coverage shall begin when the Government accepts the EVSE from the contractor FOB point of origin/destination.

Labor, parts, shipping cost, per diem and travel for warranted repairs to and from the unit's fielded location shall be the responsibility of the contractor. This shall include 3rd party Original Equipment Manufacturer (OEM) selected Offerors for corrective action or warranty issues on the EVSE if applicable.

The offeror shall provide a toll free 24 hour / 7 day call center customer care access for EVSE and data access systems. The offeror shall be equipped to address EVSE hardware issues and network/connectivity issues both remotely (via phone or internet) and on site by a qualified technician, as needed. The offeror shall provide detailed information describing their customer service access for the EVSE.

4.5.7 Option FFPS Forward Facing Passenger Seating

If option FFPS is selected, the contractor shall provide forward facing passenger seating equipped with seatbelts.

The seats shall be, American Seating Insight Prime +, or equal, thermoplastic seats with stainless steel frame, two-tone inserts, grab rail, drain hole, (meeting FMVSS 302 and FTA Docket 90-A). The color of the seats shall be pewter grey. The offeror shall furnish a seating layout drawing which will be approved by the Government.

The seating layout, make/model, and color shall be reviewed and approved by the government.

As applicable, hip-to-nee room measured from the front of one seat back horizontally across the highest part of the seat to the seat or panel immediately in front shall be no less than 26 inches. At all seating positions in paired transverse seats immediately behind other seating positions, hip-to-knee room shall be no less than 26 inches. Seat height, except for rear bench seat, shall be 17 inches. The rear bench seating height shall not exceed 21 inches.

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4.5.8 Option VSS Video Surveillance System

A functional video surveillance system shall be provided and shall include the following as a minimum:

Eight (8) high-quality, durable video cameras with built-in microphones,

Six (6) cameras shall be installed inside the bus and two (2) cameras shall be installed outside the bus. Specific camera locations shall be determined after contract award by the Government in consultation with the Contractor.

The cameras shall be hardwired, not wireless

Each camera shall be high definition (HD) (minimum of 720 pixels)

Each camera shall have true day/night (TDN) imagery and infrared capabilities

Each camera shall be capable of capturing full-color images

The mobile recording device shall be a standalone device. The recording device- digital video recorder shall not be equipped with networking capabilities, shall not be equipped with wireless connectivity (Wi-Fi, Bluetooth, Streaming, etc....), and shall not utilize cloud storage services.

The mobile recording device shall have a minimum of 12 audio and video inputs and 2 audio and video outputs

The mobile recording device shall have video and audio compression capabilities

The mobile recording device shall be able to record high-definition (HD) (minimum of 720 pixels), audible, full-color video. The viewer shall be able to zoom in on a specific location and clearly see high-definition (HD) video and hear all sound

The mobile recording device shall be capable of storing at least 2 TB of data

The mobile recording device shall incorporate a secondary or redundant storage option, such as an SD card

The mobile recording device shall include UPS battery backup to ensure successful system shutdown in the event of power loss or if the bus is immobilized prior to service

The mobile recording device shall include a system status module with integrated panic button. The status module should display and mimic DVR status indicator light as well as have RCA video input for system programming

Cameras and mobile recording device shall be rated for shock and vibration

Video surveillance equipment, including, but not limited to, cameras and mobile recording devices, shall be durable and able to operate in specified conditions